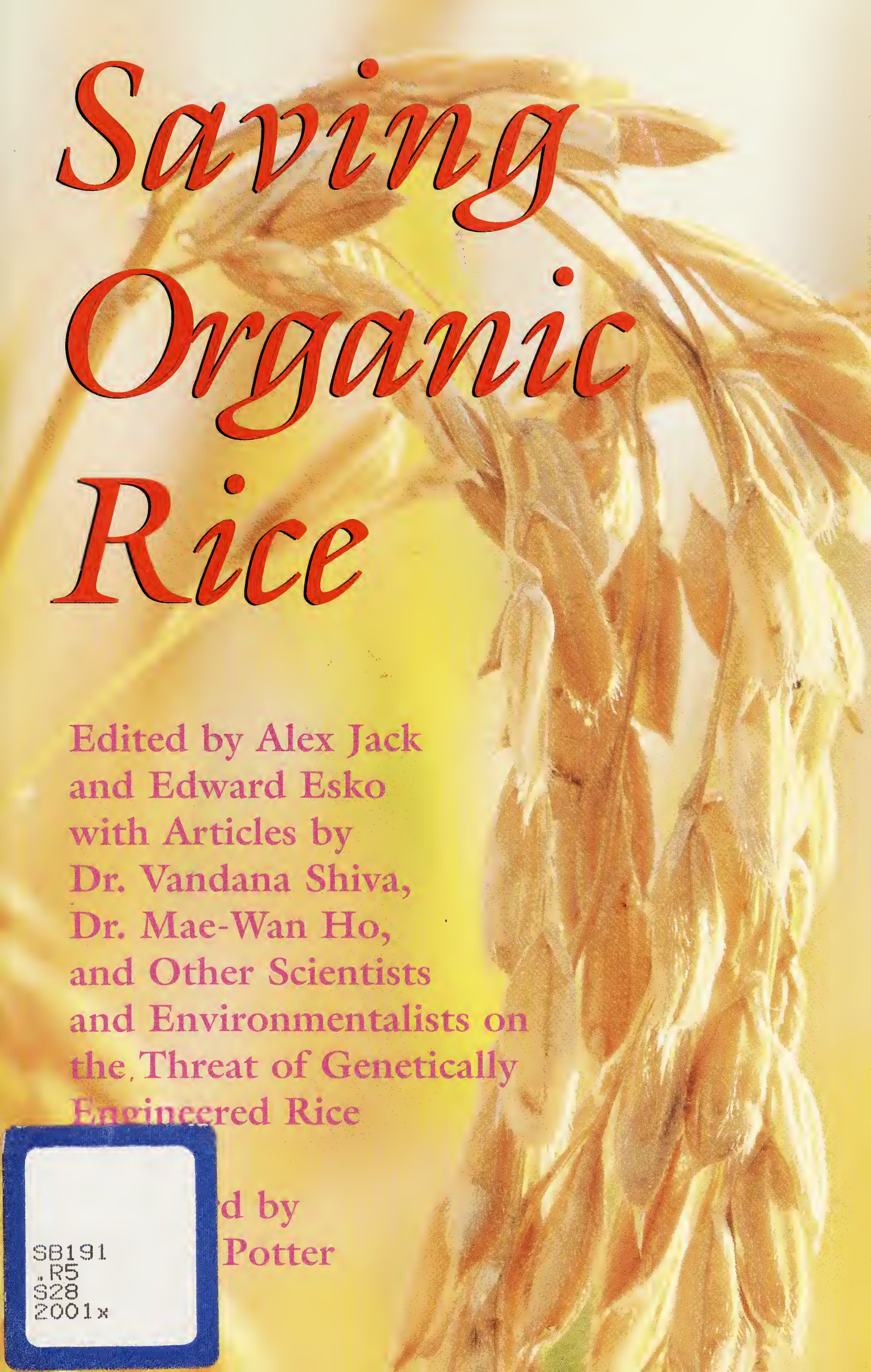




Saving Organic Rice

Edited by Alex Jack
and Edward Esko
with Articles by
Dr. Vandana Shiva,
Dr. Mae-Wan Ho,
and Other Scientists
and Environmentalists on
the Threat of Genetically
Engineered Rice

ed by
Potter

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SAVING ORGANIC RICE

*The Threat of Genetically-Engineered Rice
to Organic Farming, World Health,
and the Future of Humanity*

Edited by Alex Jack and Edward Esko
with Articles by Vandana Shiva,
Mae-Wan Ho, and Other
Scientists and Environmentalists

Foreword by Michael Potter,
President, Eden Foods

Amberwaves

Massachusetts • Michigan • California

*To the Spiritual Guardians, Prophets, & Troubadours of
Humanity Who Have Introduced the Culinary &
Healing Arts & Sung the Praises of Natural
Rice, Wheat, or Other Whole Grains
& Whose Spirits Reflect
God's Shining
Grace*

Saving Organic Rice

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Photo Credits: *Front Cover:* courtesy of Lundberg Family Farms, © Paul Bousquet, Box 3646, Boulder, CO 80307, (303) 499-5161. *Page 9:* Rice growing in the Sacramento Valley of California, California Rice Commission. *Page 39:* Edward Esko and Grant Lundberg by Alex Jack. *Back cover:* Sushi drawing by Palma O'Sullivan.

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Ode to Rice

"O beautiful for spacious skies, for amber waves of grain . . ."

—AMERICA THE BEAUTIFUL

"And God said, Behold I have given you every herb-bearing seed . . . to you it shall be for meat."—BOOK OF GENESIS

"All beings are evolved from food."—BHAGAVAD GITA

"After Siddhartha had cleansed himself in the River Neranjara, he took forty-nine mouthfuls of soft rice. After he had the food, he placed the gold cup afloat in the river, making the solemn resolution: 'If I shall become a Buddha today, may this gold cup float upstream.' The gold cup floated up stream for eighty cubits and then sank down."—LIFE OF BUDDHA

"They had best not stir the rice, though it sticks to the pot."

—Miguel de Cervantes, DON QUIXOTE

"When I was about 16 years of age I happened to meet with a book, written by one Tryon, recommending a vegetable diet. I determined to go into it. . . . I made myself acquainted with Tryon's manner of preparing some of his dishes, such as boiling . . . rice . . . I made the greater progress, from that greater clearness of head and quicker apprehension which usually attend temperance in eating and drinking."—BENJAMIN FRANKLIN, AUTOBIOGRAPHY

The Arcadia region of South Carolina "looked like fairyland . . . [it] stands on a sand hill, high for the Country, with [its] Rice fields below; the contrast of which with the lands back of it . . . is scarcely to be conceived."—GEORGE WASHINGTON

"All living things are seen to have spirit, but rice alone is said to have a soul. Djiwa, the soul, lives happily in the fields where rice grows, but has to be enticed to stay with the crop after harvesting."—LYALL WATSON, GIFTS OF UNKNOWN THINGS

"The simplest dishes are the hardest to make. The highest art in cooking is the preparation of a bowl of rice."—MICHIO KUSHI

Foreword

"By his activity in this fight he brought down on his head the lasting hatred of all the food adulterators in America. Until this day no name is more abhorrent to them than that of Theodore Roosevelt. And that organization of food adulterators is one of the most powerful political influences the country has ever had to deal with. It has openly defied the national and state governments for a quarter of a century. The enormous profits it realizes from the processes of imitation and adulteration permit it to have a mighty barrel to spend in influencing legislation . . . "

—"THEODORE ROOSEVELT AND PURE FOODS," *NATIONAL FOOD MAGAZINE*, SEPTEMBER 1912

The more things change, the more they stay the same.

Today we face a far more permanent adulteration of our food, the pollution of our food gene pool by giant corporations which own this DNA and are only interested in profit. This affront to organic agriculture and enormous burden on companies like Eden Foods to acquire non-genetically polluted organic food, is the most serious threat to the freedom of humanity from corporate exploitation ever. It strikes me as ironic that food would be genetically engineered to allow the killing of all life in the soil with toxic herbicides, when the inherent nature of anything considered food is to nurture life.

These same corporations are, today, actively working to acquire ownership of all the fresh water in India and Mexico, with plans for owning all fresh water.

Nature created perfect food. It is arrogant to think that a scientist can make food better by manipulating DNA. It is far more likely that this tampering will limit the expression of the genetic potentialities of life. Reck-

Michael Potter, chairman and president of Eden Foods, is a pioneer in the organic, natural foods industry. Since 1993, Eden has had a GEO-free policy and works actively with farmers and suppliers to ensure the quality of their crops, including comprehensive in-house GEO testing of each batch of corn and soybeans.

lessly releasing new man-made, untested organisms into our environment may lead to surprising, devastating results, and has already caused enormous economic damage.

Genetic science is not itself bad. It has potential for creating new silver bullet medicines of value. Tampering with the DNA in our food supply is another thing, and the overwhelming majority of the people on Earth do not want it done.

I have stood on huge tracts of dead soil, all life choked out of it by agricultural chemicals made by these same corporations. For Nature to repair this requires thousands of years; but, we are “the salt of the Earth,” and the magic of man is that we can repair this damage using organic techniques in three to five years. What power we have to nurture life, or kill, if we do not act wisely and prevent further processes of imitation and adulteration of food.

Michael Potter
Clinton, Michigan
February 8, 2001

Introduction

“If current trends in the genetic engineering of California’s major export crops continue, the state’s producers could be facing gathering storm clouds in their export markets and the potential collapse of its organic food industry.”—CALIFORNIA AT THE CROSSROADS, GREENPEACE REPORT ON GE RICE AND OTHER CROPS, 2000

“The greatest service which can be rendered any country is, to add an useful plant to its culture; especially, a grain.”

—THOMAS JEFFERSON, ON INTRODUCING DRY LAND RICE TO AMERICA

In 2001, genetically engineered (GE) rice will be introduced in the U.S., China, India, Japan, and other countries. Whole grains are the foundation of human existence. The introduction of new GE cereal grains imperils millions of years of natural evolution and represents an unprecedented threat to the survival of our species and its diverse cultures and civilizations.

The biotech companies attack natural and organic brown rice as “nutritionally deficient” and plan to replace it with “improved” GE varieties that are designed to be milled into 100% white rice. They have set a goal of converting one third to one half of the rice fields in California’s Sacramento Valley to growing GE varieties. This is the site of organic farms that produce 80% of the brown rice in America. GE rice will soon be introduced in Arkansas, Texas, and other Southern areas where organic brown rice is also grown. Unless we join together and act now, it is only a matter of time before heirloom varieties disappear and GE rice pollen contaminates natural rice fields, threatening the heart of the organic foods movement.

Following the StarLink corn fiasco this past year, up to 50% of the corn in Iowa and some regions registered positive for GE organisms though less than 1% had been planted. The same thing could easily happen with rice, wheat, and other crops. “Our investigations thus far from the 2000 harvest lead us to believe that virtually all of the seed corn in the United States is contaminated with at least a trace of GE material,” warns David Gould, a member of the certification committee of the California Certified Organic Farmers. “Even the organic lots are showing traces of biotech varieties.”

Over 200 years ago, America faced a similar catastrophe. During the Revolutionary War, the British seized all the seed rice in the Carolinas, threatening the South's major food crop. While ambassador to France, Thomas Jefferson journeyed over the Alps to Italy in 1787 to locate a new supply. The local authorities forbade the export of rice in the hull, under penalty of death, but Jefferson smuggled several large handfuls in his coat pockets, and rice farming in America was saved. The spirit of liberty, justice, and freedom represented by Jefferson, Franklin (who began eating rice when he adopted a vegetable diet in his teens), George Washington (who enjoyed rice and said the beauty of natural rice fields "is scarcely to be conceived"), Thoreau, Gandhi, George and Lima Ohsawa, and other great teachers of humanity must be reawakened as the GE peril approaches.

In the face of this threat, we are forming Amberwaves, a planetary network of individuals, families, and organizations, to help preserve organic rice and other essential foods for future generations. This book is the first step in a campaign to awaken the public to this central issue. We invite you to participate actively in the Amberwaves network. Practical steps are listed on the back page of the book, including a Save Organic Rice Petition and a GE-Rice Free Zone Declaration.

In compiling this volume, we are grateful to Michio and Aveline Kushi, who have pioneered the introduction of organic brown rice in America; the visionary authors whose articles are included in this volume; and countless mothers, fathers, children, grandparents, and other concerned citizens who are making their voices heard at the grassroots level. We deeply appreciate the enthusiastic support of America's organic rice farmers, including Grant Lundberg and Sabrina Wootton of Lundberg Family Farms, Willadean and Ron Hogue of Southern Brown Rice, and Donna Wallace of Bayles Ranch. Cornelia and Jiro Aihara of the Vega Study Center; Carl Ferré of the George Ohsawa Macrobiotic Foundation; and Michael Potter, president of Eden Foods, have also actively assisted our efforts for which we are thankful. We are grateful to our spouses, Gale and Wendy, who have contributed to the spread of delicious, healthful, natural foods cooking; our families; and to our students and colleagues across the country and around the world.

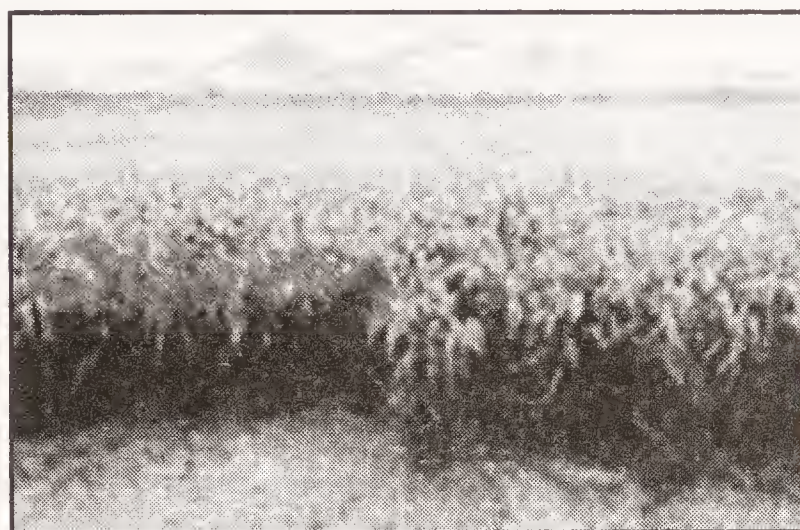
In early 2001, as this book went to press, the U.S. Department of Agriculture announced that the first two GE rice varieties entered commercial production. Monsanto began the first trials of GE wheat in four states, and the rice genome was mapped in what biotech companies hailed as a breakthrough in the "improvement" of the world's major food crop. Amber waves or altered waves? The choice is up to us.

Alex Jack and Edward Esko
Becket, Mass. and Clinton, Mich.
February 4, 2001

1

Saving Brown Rice

By Alex Jack



Rice growing in California's Sacramento Valley.

"Genetically modified (GM) rice is coming to California. It will be introduced in the U.S. rice industry as early as 2001. . . . Early estimates anticipate significant use of herbicide-resistant rice varieties in California with as much as one-third to one-half of planted acres in GE rice in 10 years."

—TIMOTHY JOHNSON, EXECUTIVE DIRECTOR, CALIFORNIA RICE COMMISSION

During the 1960s, the world was awakened to the suffering of the war in Vietnam by the self-immolation of Buddhist monks and nuns. The Buddhist movement, which sought a neutral Southeast Asia independent of both the Eastern and Western blocs, gathered widespread popular support. As a re-

Alex Jack is an author, teacher, and dietary counselor. Over the last 25 years, he has served as editor-in-chief of the East West Journal, general manager of the Kushi Institute, and director of the One Peaceful World Society. His books include The Cancer Prevention Diet (with Michio Kushi), Amber Waves of Grain (with Gale Jack), The Mozart Effect (with Don Campbell), Let Food Be Thy Medicine, and Imagine a World Without Monarch Butterflies.

porter covering the war, I had an opportunity to interview Thich Tri Quang, the charismatic Zen master who led the peace movement and who had taken sanctuary in the An Quang Pagoda near Saigon.

Heavily armed police and military forces ringed the temple, but even they dared not enter and arrest Thich Tri Quang because of his undisputed spiritual authority. Thich Tri Quang was slight and soft-spoken, but radiated a commanding presence. His eyes were positively aglow with understanding and compassion, and it was easy to see why he was considered the Gandhi of his country. I asked him many questions about war and peace, and he patiently explained the goals of the Third Force movement and delved into many areas of Far Eastern history and philosophy. I mentioned that I had just come back from the Gulf of Tonkin where I had interviewed American pilots on an aircraft carrier bombing North Vietnam. They confided that nuclear weapons were secretly stored aboard the ship.

"Aren't you afraid that the war in Southeast Asia could escalate into a nuclear conflict?" I inquired.

Thich Tri Quang replied that he had heard similar reports that the Soviets had smuggled nuclear devices into Hanoi. "Of course, it would be a great tragedy if nuclear war broke out," he replied. "But there is something far worse than nuclear war."

I was shocked by his reply and time seemed to stand still. I could not conceive of anything worse than nuclear war. As a child in the 1950s, I had met victims of the atomic bombing of Hiroshima on a trip to Japan with my father, a social activist, and had grown up believing that World War III would be the end of human existence.

"What could be worse than nuclear war?" I asked.

"The rice," Thich Tri Quang replied simply.

Again I was puzzled. What did rice have to do with the end of humanity? I was rendered speechless by this Zen koan, or paradoxical riddle.

"Rice is the foundation of culture and civilization in the Far East," Thich Tri Quang went on to observe. "It nourishes personal health and family life, art and culture, philosophy and religion. During the war, most of the rice fields here have been destroyed by bombs and chemicals. The rice in our country now comes primarily from Texas. It is a polished white rice—completely devoid of life and energy."

I knew the difference between brown and white rice. I had recently become vegetarian and was eating whole grain rice and other wholesome foods whenever possible. But I had never connected food with social health or world peace.

"Vietnam has been at war for nearly 2000 years," Thich Tri Quang reflected. "We are very good at survival. If the North wins, human rights will be jeopardized, but we will survive. If the South wins, big corporations will take over the economy, but we will survive. If nuclear war breaks out, we

are a small agricultural country and will also survive. But if rice perishes and Vietnam adopts the modern way of eating, it will be the true end of our civilization and culture.”

At the time, I did not fully understand Thich Tri Quang’s perspective. But after returning home, I gradually came to see that our daily food has a profound social and environmental impact. Over the last thirty years, I have been active in the natural health community as a journalist, macrobiotic teacher and counselor, and parent. During this time, a plant-centered diet moved from the counterculture into the mainstream. Brown rice, tofu, and other organic foods once identified with Zen beatniks and hippies are now eaten by millions of people. The U.S. government’s latest dietary guidelines call for the American people to eat brown rice and other whole grains every day as the center of the meal!

As the new century begins, it is increasingly clear that humanity is splitting into two directions: the artificial and the natural. The competition between GE food and organic food lies at the heart of this conflict. Biotechnology has targeted the Sacramento Valley as ground zero in its campaign to take over control of the world’s food supply, and GE rice will be its poster crop. An epic drama could unfold in the next few years in the rice fields of northern California (as well as the organic vegetable farms of the Central Valley), eerily reminiscent of the war in Vietnam a generation ago! Thich Tri Quang’s concern about the fate of humanity being intertwined with the fate of natural rice and other essential foods has become tragically prophetic.

The Future of Humanity

“You must then watch for the world, gird up your loins with all your might lest the robbers find a way to take advantage of you, because they will find the weakness which you (least) expect. Let there be among you a person of wisdom. When the grain ripened, the reaper came in haste with his sickle in his hand and he harvested it. Whoever has ears to hear, let him hear.”—JESUS, GOSPEL OF THOMAS

The introduction of genetically engineered (GE) rice in America, China, Japan, India, and other parts of the world is imminent. Several dozen varieties of artificial rice are in development, and the biotech companies are competing to release them over the next few years. Three billion people—one half the world’s population—eat rice daily. Rice and other whole grains represent the center of the spiral of human evolution on this planet for millions of years. The future of humanity depends on preserving the natural quality of rice and other grains.

Until now, the biotech agriculture industry has focused on designing genetically engineered crops that are used primarily to feed livestock. Within the last few years, about 50% of America's soybeans, 35% of its corn, and 40% of its cotton have been genetically altered. Rice, wheat, and other grains that are produced for direct human consumption are not in the process of being genetically engineered and will soon be available.

Like other GE foods, the long-term effects of altered rice on human health and the environment have not been tested. Since no labeling is required, consumers in the United States have no way of knowing what they are buying and eating or the potential hazards to their families. Natural rice currently produced in America and other countries (both organically and conventionally grown) would also be at risk from "genetic pollution," the drift of pollen from nearby GE rice fields that could contaminate ordinary crops. Multinational corporations are also patenting basmati rice, jasmine rice, and other traditional strains in order to obtain a monopoly on the use of their names and sale under new international trade regulations, further imperiling the present supply of natural rice. Already there is a growing body of scientific and medical evidence on the dangers and possible future hazards of GE foods.

Over the last year, about a dozen varieties of new artificial rice have been reported in development several traditional varieties have been patented:

- **Italy:** Scientists at the Catholic University of Piacenza announced that they are developing a GE rice spliced with a gene from soybeans that resists a parasitic insect that is common in paddy fields in the south of France and in China.
- **Europe and Japan:** Biotech industry officials predicted that Aventis SA, a large Franco-German life sciences company, would be the first biotech firm to begin commercial production of GE rice in the United States and South America in 2001. Aventis announced its intention to apply to Japan's Health Ministry in mid-2000 for approval to import its herbicide-resistant varieties of GE rice to Japan by the end of 2001.
- **United States:** Monsanto, the leading biotech company, announced that it was developing GE rice in China and India and that it would introduce a GE rice in the United States as early as 2002.
- **Switzerland:** Scientists reported that they have developed a new genetically modified "golden rice" that is higher in beta carotene. The rice is designed to supplement diets low in vitamin A and protect Asian children from blindness. Another strain is being developed to

reduce iron-deficiency anemia.

- **China:** Xinhua News Agency reported that scientists in the coastal Zhejiang Province have developed a GE strain of rice that is resistant to herbicides. In Beijing, officials announced that GE rice, corn, and soybeans would begin production in 2001. “We have to develop our own technology on GE crops ahead of the inroads of foreign GE seeds,” said Lie Depu of the Chinese Academy of Agricultural Sciences.

- **Japan:** The Agriculture Ministry announced plans to register its newest rice variety as Japan’s first domestically produced GE food as early as 2001. In another development, researchers at the Science University of Tokyo announced that they had succeeded in splicing hepatitis B antibody to rice plants.

- **Philippines:** Researchers announced the development of a new GE rice that produced 35% higher yields in field tests in China, Korea, and Chile. The International Rice Research Institute also said that it was planning to use genetic engineering to increase the nutritional value of the Indica tropical rice plants grown in South and Southeast Asia by introducing vitamin A taken from daffodil genes.

- **United States:** RiceTec, a Texas corporation, has taken out a patent on 22 basmati rice lines and grain and claimed exclusive right to use the name “Basmati” under new international trade regulations. Basmati rice, grown in India and Pakistan for centuries, is prized around the world for its fragrant aroma and distinctive taste. Each year, India alone exports \$425 million of basmati rice to the U.S. After American courts upheld the patent, the Indian government charged the U.S. with “biopiracy” and is seeking redress in international courts. U.S. courts have upheld similar patenting of common Mexican beans and other staple crops from around the world, and U.S. customs officials are actively enforcing bans on the import of traditional seeds.

- **United Kingdom:** AstraZeneca, a pharmaceutical giant, announced that within three years it would sell a GE “golden rice” in the rich countries and give it away in the poor countries.

- **Australia:** Scientists are developing a technique to insert a measles gene into GE rice as an alternative to traditional vaccination against the disease. The measles vaccination may be distributed to children in

the form of rice flour milk and is expected to be available within five years.

- **Africa:** The West Africa Rice Development Association (WARDA) has developed a new GE rice combining the African species *Oryza glaberrima* and the Asian species *Oryza sativa*, which they say will double or triple production among West Africa's 20 million rice farmers. The new rice is expected to be released in Guinea, Cote d'Ivoire, and Sierra Leone later this year.

- **United States:** Scientists at Washington State University announced in June they have successfully engineered a new strain of rice with corn genes, and preliminary field trials showed that it increased yields up to 35%. Other researchers are modifying rice at the university by targeting seeds to make larger grains.

- **South Korea:** A new GE rice that can yield up to 26% has been developed by researchers at Chonnam National University and the Scigen Harvest Co. The GE rice includes a gene from a microbe, *bacillus subtilis*, that increases photosynthesis.

The Health and Environmental Effects of Genetically Modified Rice

"Antibiotic resistance, the threat of new allergic reactions, and the unknown hazards of transgenic DNA mean that on health ground alone the impact of genetically modified organisms must be fully assessed before they are released."—BRITISH MEDICAL ASSOCIATION

Although no long-term studies have been done on the GE rice and other foods, a surprising amount of short-term research raised serious questions about the viability of new genetic technologies. In Brazil, according to the *New England Journal of Medicine*, GE soybeans containing genes spliced from nuts posed a lethal danger to people allergic to nuts and had to be recalled. In Scotland, GE potatoes fed to rats resulted in stunted growth and damage to major organs, including kidney, spleen, thymus, and stomach. The *Journal of Medicinal Foods* reported that GE soybeans have from 12-14% less phytoestrogens that help protect against cancer and heart disease. Cornell University researchers reported 44% of the larvae of monarch butterflies died after several days' exposure to pollen from GE corn. There have been several hundred such reports and studies published in peer-reviewed scientific, medical, and environmental journals in the last few

years. (See *Imagine a World without Monarch Butterflies* by Alex Jack for a summary of current scientific and medical research.)

In India, Europe, and the United States, scientists have expressed concern about the impact of new hybrid and GE varieties of rice on human health, the environment, and society:

- **Increased Vitamin A Toxicity:** Rice engineered to be higher in vitamin A could lead to excessive consumption of this nutrient and consequent vitamin A toxicity, which is associated with abdominal pain, nausea, vomiting, dizziness, bone and joint pain, hair loss, dryness and fissures of the lips, weight loss, and other symptoms. GE rice could be particularly dangerous to pregnant women, young children, people with liver disease, diabetics, and hyperthyroid sufferers.

- **Increased Toxins, Allergens, and Disease:** Glyphosate, the active ingredient in Monsanto's GE rice, is the third major cause of pesticide-related illnesses among agricultural workers. According to the Northwest Coalition for Alternatives to Pesticide Use, glyphosate can cause eye and skin irritation, headache, nausea, elevated blood pressure, and heart palpitations, and farmers exposed to glyphosate herbicides are at higher risk of miscarriages, premature birth, and non-Hodgkin's lymphoma.

- **Increased Dependence on White Rice:** The new GE "golden rice" is specifically designed to concentrate beta-carotene in the endosperm of the plant that is retained after the fiber-rich bran and germ are removed in milling. In brief, the modified rice is designed to reinforce eating nutritionally deficient polished white rice as a daily staple rather than nutritious brown rice that contains some naturally occurring carotenoids that are lost in polishing.

- **Contamination of Organic and Conventional Crops:** Like other altered crops, GE rice may release engineered pollen, seed, pesticides, toxins, bacteria, viruses, or other organisms into the environment that can crossbreed with or contaminate ordinary crops. To date, thousands of acres of organic crops, as well as numerous beehives, have been contaminated by GE corn or rapeseed pollen that has drifted up to six miles. In a report sponsored by the British government, *Organic Farming and Gene Transfer from Genetically Modified Crops*, researchers concluded that contamination was inevitable: "Once GE crops are released they, like all crops, cannot be completely contained. The same principle is true for spray or fertilizer drift from one farming system to another. There has always been the possibility of hy-

bridization and seed mixing between organic crops and conventional crops.” In Canada, many farmers have given up growing organic canola because it is too risky and they fear losing organic certification if any trace of GEOs are found in their produce. “The whole issue of genetic pollution is one of the only things that can seriously damage the organic movement—if not kill it,” says Stewart Wells, an organic farmer and Saskatchewan director of the National Farmers Union.

- **Increased Water Pollution:** The introduction of high-yielding GE rice strains will involve a shift from water conserving agricultural methods to water intensive production methods since high yielding rice varieties are highly water demanding, leading to further burden on ground water or intensive irrigation from large dams and associated problems with water-logging and salinization.

- **Threat to Monarch Butterflies and Other Wildlife:** Some new GE rice contains Bt, the engineered pesticide that is linked with killing Monarch butterflies and other beneficial insects.

- **Increased Pesticides and Chemical Inputs:** Like corn, GE rice may require more pesticides, chemical fertilizers, and other sprays. According to a former director of the National Academy of Sciences Board on Agriculture, “Clearly *Bt* corn has not reduced insecticide use, and indeed probably has and will continue to increase it.”

- **Emergence of Disease-Resistant Pests:** New strains of bacteria, viruses, and other organisms may arise. Rice containing GE Bt could produce new strains of Bt-resistant bacteria that will spread to other organic and conventional farms and crops. Naturally occurring Bt—a less toxic variety than GE Bt—is the most widely used natural pesticide. The organic foods community is worried that its industry could be destroyed if natural Bt is rendered ineffective.

- **Loss of Biodiversity:** The ecological impact on soil organisms and other organisms dependent on rice in the food chain may result in reduced biodiversity and the potential for disease and pest vulnerability.

- **Emergence of New Herbicide-Resistant Weeds:** Dr. Ren Wang, deputy director-general for research of the International Rice Research Institute and a strong supporter of biotechnology, said that attempts to develop an altered rice that is herbicide-resistant may result in the genes becoming transshifted to non-rice weed species, creating a “superweed” and that safeguards were required.

- **New Viral Diseases:** The viral promoter in “golden rice” and most other GE crops is derived from the cauliflower mosaic virus (CaMV). This promoter has a recombination hotspot that is apt to break and join with other genetic material, threatening to spread and introduce infectious viruses to other species.

- **Threat to Sustainable Agriculture:** In the first scientific audit of “Golden Rice,” the Institute of Science in Society, based in Britain, uncovered “fundamental deficiencies in all aspects, from the scientific/social rationale to the science and technology involved.” It called the new GE rice “a drain on public resources and a major obstruction to the implementation of sustainable agriculture than can provide the real solutions to world hunger and malnutrition.”

- **Unforeseen Health and Environmental Risks:** While the new GE rice strains may produce higher yields in the short run and help prevent or relief specific illnesses, they could also create unforeseen health or environmental risks whose impact may not be measured until many years or generations later. In spring 2000, Sri Lanka’s Food Advisory Committee instructed customs to prevent importation of GE foods because of long-time health and environmental implications.

Spliced Rice

“A handful of rice is worth more than a handful of gold.”

—VIETNAMESE PROVERB

Scientists in Switzerland recently announced that they have developed “golden rice,” a GE rice that could help alleviate blindness and other ailments, especially among millions of children worldwide who eat diets deficient in vitamin A. The scientists developed the new spliced rice by inserting three genes into rice that allow the plant to produce beta-carotene, a precursor to vitamin A. The GE rice will be made freely available to farmers in developing countries.

Unlike the first generation of GE foods such as soybeans, corn, and cotton that were sold in combination with GE herbicides and intended to reap large profits for the biotech food industry, this product appears to be motivated by humanitarian concerns. However, it does not address the true, underlying causes of poverty, hunger, and disease in the world today. Vitamin A deficiency, malnutrition, and the spread of infectious diseases are primari-

ly the result of modern social, economic, and political forces that have uprooted people from their farmlands and forced them into cities, slums, and marginal lands, creating a vicious cycle of hunger, disease, and destitution.

Instead of continuing to produce whole grains, vegetables, and a variety of other fresh, wholesome foods, traditional farmlands have increasingly been transformed into pasture for cattle to produce beef or into large factory farms or plantations to produce coffee, bananas, or other commodity crops for export to affluent countries. A generation ago, the Green Revolution introduced higher yielding varieties of hybrid “miracle rice” and other crops in the Third World. The result was a disaster. In India, 30,000 varieties of rice flourished until just a few years ago. Today, only 12 varieties are widely cultivated. The Bhopal disaster in which 3000 people were killed by toxic fumes from a pesticide plant was part of the Green Revolution.

Into this broken fabric, the United Nations and other well meaning social agencies arrive, dispensing tons of surplus powdered milk, sugar, white flour, and other high calorie foods that relieve starvation, but ultimately increase human illness and prolong suffering. Now GE rice is about to be added to this misguided humanitarian menu.

Despite all the hype about alleviating blindness and reducing world hunger, critics of “golden rice” have pointed out that it will supply only 4.4% of the RDA for vitamin A. “Nutrition science . . . suggests that golden rice alone will not greatly diminish vitamin A deficiency and associated blindness,” Dr. Marion Nestle, chairman of the department of nutrition and food studies at New York University, warns. “Conversion of beta carotene to vitamin A, and transport in the body to the tissues that use vitamin A, require diets adequate in fat and protein. People whose diets lack these nutrients or who have intestinal diarrheal diseases—common in developing countries—cannot obtain vitamin A from golden rice.”

Rather than put our faith in quick fixes, we should mount an effort to preserve heirloom strains of rice and other grains, beans, vegetables, fruits, and other plants that have sustained and nourished the human family from time immemorial. This is humanity’s true wealth and heritage, the richness and diversity of its traditional foods and cultures.

Whole brown rice contains a variety of nutrients, including naturally occurring carotenoids that are lost with polishing. The new spliced rice is genetically engineered to concentrate beta-carotene in the endosperm of the plant that is retained after the fiber-rich bran and germ are removed in milling. Spliced rice is designed to reinforce eating nutritionally deficient polished white rice as a daily staple rather than nutritious brown rice.

Rice that is genetically engineered to alleviate vitamin A deficiency should not be allowed to enter the human food supply. “The GM vitamin A rice is now being used as a Trojan horse to push GM crops and foods,” warns Dr. Shiva. Rather, an intensive international effort should be

launched to ensure that all of the children of the world have adequate nutrition from kale, carrots, broccoli, parsley, mustard greens, pumpkins, sweet potatoes, green peas, Brussels sprouts, and other natural sources of carotenoid-rich vegetables, as well as from whole rice and other whole grains, and other plant foods that are part of a balanced diet. Reliance on genetic engineering to relieve blindness caused by naturally correctable dietary deficiency represents a tragic loss of planetary vision.

Sustainable Solutions

*The seed of all song
is the farmer's busy hum
As he plants his rice.*
—BASHO

Contrary to the claims of the biotech industry, simple, safe natural methods are available to maintain and secure optimal human health, feed the hungry, boost yields, and protect the environment. Throughout the Third World, farmers have mobilized against GE rice, and community projects have shown that sustainable agriculture methods are readily available:

- **Multiculture Produces Higher Yields in China:** In China, planting a blend of rice varieties resulted in higher yields than planting just one variety, and farmers no longer had to spray harmful fungicides. The experiment began in 1998 when farmers in five townships in Yunnan Province switched from planting a single variety of sticky rice to alternating rows of sticky rice with other varieties. The next year, farmers in 10 townships joined the experiment, bringing the area under cultivation to 8,255 acres. Total yields increased by 89% and incidence of rice blast, a fungal disease, decreased by 94%. The experiment showed that multiculture could successfully be applied to growing rice, and that chemical sprays and GE varieties were not needed.
- **Philippine Farmers Favor Traditional Over GE Rice:** In the Philippines, rice farmers protested a plan by the International Rice Research Institute, the world's largest rice foundation, to perform field tests of GE bacterial blight rice (BB-rice). The farmers warned that the entry of GE rice will further deplete traditional rice strains that are sustainable and lead to monopoly control of rice seeds by multinational companies. "We planted IR-8 and threw away our traditional seeds," Pedring Pangan, a farmer from Calauan, recalled when the IRRI introduced "miracle rice." "We had a good first harvest, after that, we har-

vested almost nothing.” IR-8 caused brown planthopper, a deadly pest, to proliferate. Farmer Tata Gonying Velasco said that the new varieties have not only eroded the diversity of traditional rice but also wiped out much of the farmers’ history as “stewards of the seeds.” MASIPAG, a network of scientists and farmers promoting sustainable agriculture, supports six cultural management practices to control bacterial blight instead of planting BB-rice. These include low use of nitrogen fertilizer, adequate irrigation and drainage, seedbanking of blight resistant plants, and maintenance of crop diversity as well as appropriate transplanting and proper disposal of infected plants.

- **Thai Farmers Laud Traditional Varieties:** In Thailand, farmers from around the country flocked to a Rice Forum held near Bangkok to protest the introduction of GE rice. Dr. Ampon Kittiampon, Deputy Secretary of the Ministry of Agriculture and Cooperation, decried the emphasis on cost-effectiveness over societal values. Joni, leader of the Karen tribes, said that “rice is life for the Karen” and that losing the seed to biotech varieties would be to lose life itself. One hundred traditional varieties have declined to only five in recent years. Dayene Siripetra from the Rice Spirit Foundation said that the Green Revolution introduced to Thailand by the World Bank and the Rockefeller Foundation caused drastic loss of traditional varieties through emphasis on high yield and high input. He said that farmers were encouraged to exchange traditional varieties for new ones that turned out to be susceptible to disease. The consensus of Thai farmers was that the use of pesticides and fertilizers resulted in many diseases, while traditional varieties yielded healthier, more fragrant varieties. After the forum, the Rice Research Institute in Thailand announced that they would put all GE rice in the gene bank but not carry out any more research at this time.

- **Bangladesh Home Gardens Supply Vitamin A:** In Bangladesh, traditional cultivation methods and the introduction of small home gardens with vegetables and fruits high in vitamins have dramatically decreased vitamin A deficiency (VAD). Landless families without any land were enabled to grow vines on their houses and plant beans, pumpkins and bottle-gourds in the vines which contain edible leaves high in vitamin A. Since 1998, 3 million people have participated in the project.

- **Solar Drying Techniques Perfected in Africa:** In Mali, Burkina Faso, Niger, and other African countries, solar drying methods have been developed to conserve the nutrients in traditional vegetables and fruits for up to 6 months, preventing vitamin A deficiency.

- **New Organic Methods Boost Rice Harvest in Madagascar:** In Madagascar, small rice farmers increased yields 4 to 5 times by developing new organic techniques known as the System of Rice Intensification (SRI). Instead of transplanting groups of seedlings into fields at about 8 weeks, seedlings are transplanted at about six days and planted individually. Instead of keeping the plants constantly submerged in water, the rice is kept wet only during the reproductive stage. It is irrigated in the evening and allowed to remain dry during the day. Typical harvests of 2 tons per hectare increased to 8 to 10 tons per hectare without chemical fertilizers, pesticides, or expensive seed varieties. SRI has since spread to Bangladesh, China, and Indonesia.

Special Report from the Sacramento Valley

“If you wish to acquire a more definite physical feeling for what’s meant by ‘spirit,’ just walk through a field of grain—for example, rice, wheat, or corn. That sudden sweep of energy which you feel rising up on all sides is the vertical polarization of the growing grain. This earth/heaven orientation of energy is not destroyed by the chemical processes of cooking or digestion and goes to form our very cells, including, of course, our central nervous system.”—MICHIO KUSHI

Last summer my colleague Edward Esko and I visited the Sacramento Valley of California where most of the organic brown rice, and about 25% of all American rice, is planted and met with rice farmers. The valley is ringed by mountains on three sides and the crops are nourished by fresh mountain water from the nearby high Sierra range. The rice fields themselves were beautiful, radiating strong, peaceful energy and teeming with sparrows, dragonflies, tiny water creatures, and other forms of life.

We met with Grant Lundberg, CEO of Lundberg Family Farms, the principal supplier of organic brown rice in America and a pioneer critic of genetic engineering in the natural foods industry. He compared the nation’s policy of handling GE corn and soybeans to “a train wreck.” He said that most of the rice farmers in the valley were opposed to GEOs and concerned that future contamination of their crops could dry up their markets. Most of the rice grown in California is short- and medium-grain Japonica rice, and 40 percent of it is exported to Japan, where surveys indicate that consumers are strongly opposed to biotech food. The first GE rice in America, he predicted, would be introduced in Texas, Louisiana, Arkansas, or other region in the South, where long-grain Indica strains are grown. These account for about 75% of U.S. production and are primarily milled into white rice, converted rice, and other highly processed varieties.

At a recent Annual Rice Growers Meetings in Colusa and Yuba City, California, the rice industry was told that it was not a question *if* GE varieties of rice would be introduced in the Valley, but *when*. Monsanto is developing Roundup Ready Rice, similar to Roundup Ready Corn and Roundup Ready Cotton. AgrEvo, another large biotech company, is developing Liberty Link Rice, which also includes an altered bacteria that is tolerant to chemical herbicides sprayed on the rice fields. Imi-rice, the third type, is designed to resist imidazilinone-type herbicides. The California Rice Commission predicts that the first GE variety to be introduced in California will be an herbicide-resistant medium-grain variety known as M202. In the Southern states, it predicts the first strain will be a variety known as Bengal.

In her address to the rice growers, researcher Peggy G. Lemaux, a strong proponent of the new biotech varieties of rice, conceded that outcrossing with conventional rice varieties is a potential problem. When panicles are touching, she explained, outcrossing frequencies can be as high as 50 percent. Separating plants 10 to 20 feet reduces outcrossing substantially. Herbicides could also drift from one farm to another, she noted. "Since there are no outward signs that would alert one grower to the varieties in his neighbors' fields this could lead to a serious problem with drift damage and it is not clear how this could be managed."

In the face of this threat, Grant explained, the state was considering enacting the California Rice Certification Act of 2000 to ensure that different varieties of rice were not inadvertently commingled. Though the Act does not mention genetic engineering, it was clearly directed at segregating GE and non-GE varieties and was strongly opposed by the biotech industry. The California rice industry sponsored the Act with an eye to implementing it by April, 2001, when Japan officially requires labeling of all imported GE crops. Meanwhile, Turkey, the second major importer of California rice, had just been approved for entry into the European Union and would probably require GE labeling like other EU countries.

The GE Roundup and Liberty rice lines are currently being tested in California at the Rice Experiment Station, on private rice paddies, and under research programs at the University of California, Davis. The nation's first GE food, the Flavr-Savr Tomato, was developed at UC, Davis, just north of Sacramento, and one-third of all the world's biotech companies were founded by UC faculty members. Several of the approximately 200 biotech experiments at UC-Davis have been the recent target of crop resisters—militant environmentalists who have destroyed GE crops. Several of the organic farmers we talked to reported that they had heard rumors of secret test plots of GM-rice in the Sacramento Valley. With several quaint World War I-like bi-planes (used to broadcast seed and monitor the growing green shoots) circling lazily over the rice fields, there was an exotic James Bond aura to the whole situation.

One of the most chilling scenarios involves the use of GE rice to produce chemicals for the pharmaceutical industry rather than as food. In 1997, the California Rice Growers Association concluded a business deal with Applied Phytologics, Inc. (API) to grow GE rice spliced with different industrial and pharmaceutical chemicals, including three human proteins: anti-thrombin, anti-trypsin, and serum albumin. “There are significant human health concerns about the production of human pharmaceutical proteins and other industrial chemicals, like detergent enzymes, in rice,” explains Dr. Doreen Stabinsky in “California at the Crossroads,” a report on the impact of genetic engineering on the state’s agriculture. “Clearly such rice should not be eaten by humans. Yet there are many ways that rice intended for human consumption could be contaminated by drug-containing rice—starting from seed contamination in the hull of the airplanes used to fly seed onto fields, to mixing of the rice harvest in trucks, dryers, and mills.” Fortunately, the rice association ended the agreement for undisclosed reasons, but API continues to contract with independent growers in the Sacramento Valley.

Overall, the California rice industry is more savvy and politically aware than other segments of modern agriculture. In a position paper published in the *Rice Journal*, Timothy Johnson, the executive director of the California Rice Commission, presented a balanced account of the subject, summarizing objections to GE rice and growing worldwide opposition, as well as potential high-tech benefits. He explained that the California Rice Commission has developed a pragmatic approach to the GE issue designed “to meet customer demand while supporting ongoing research into GEO technology.” Elements of the policy include:

- continued support of GEO research
- support of commercialization and marketing efforts when markets exist, research shows a new product to be safe, and agreement over the environmental fate of genetic material is reached
- support for current FDA regulations and the development of voluntary labeling guidelines
- identity preservation to be developed to ensure the segregation of GE and non-GE rice
- investigate the development of a program to ensure seed and crop quality

“The view on the horizon could be promising for GE crops,” he concluded, “especially those with significant nutritional and other end-user benefits. The California rice industry intends to proceed down the path toward full integration of GE products in a deliberate and controlled manner. The successful introduction of GE rice will benefit growers and marketers while

maintaining important high-value markets for California rice. And, by proceeding in a prudent manner, the California rice industry can ensure its ability to meet the demands of its customers.”

Last fall, the California legislature passed the Rice Certification Act mentioned above, and Governor Gray Davis signed it into law. The Act would protect California rice farmers from some of the economic chaos and hardship corn and soy farmers experienced after GE crops were introduced, but it does not address the major health and environmental issues, such as the creation of allergens, development of herbicide-resistant weeds, and, most importantly, the contamination of natural and organic fields by GEOs.

The biotech industry has vowed to continue with its plans to introduce GE rice and other grains. The Rice Millers’ Association and U.S. Rice Producers’ Group, two of the nation’s largest trade groups, staunchly support biotechnology and oppose consumer labeling. Ann Veneman, a California native and the new secretary of agriculture in the George W. Bush administration, is a strong advocate of biotechnology. As former Secretary of the California Department of Food and Agriculture, she stated, “The United States is the world leader in developing new agricultural products from biotechnology. The potential benefits of this new technology for producers, consumers, and the environment are enormous.” A long-time resident of Sacramento, she served on the board of Calgene, the developer of the Flavr-Savr tomato, before the company was taken over by Monsanto. At a meeting of biotech officials this past fall to plan the introduction of GE crops in Monterey county, she alluded to GE “golden rice” in her address, “We simply will not be able to feed the world without biotechnology.”

Unless public opinion mobilizes in favor of a moratorium on new GE varieties, it is only a matter of time before GE rice pollen wafts through the majestic Sacramento Valley, imperiling the intricate ecosystem that has developed naturally over millions of years and threatening the heart of the organic foods movement in America. Altogether, nearly 200 different species of animals live in California rice fields, including 141 species of birds, 28 mammals, and 24 amphibians and reptiles. Thirty of these species are listed as threatened or endangered. In the late autumn and winter, the rice wetlands host 5 million Canadian ducks and geese on their annual migration south. No other crop in the world provides benefit for so much wildlife as natural rice. As we took leave of the bright green California rice fields, a beautiful orange-and-black Monarch butterfly fluttered by, reminding us of the underlying harmony of nature and the urgency of our task.

The Campaign to Label GE Foods

“It is false to pretend that there is any distance which is going to prevent some

contamination.”—MICHAEL MEACHER, ENVIRONMENT MINISTER OF THE UNITED KINGDOM, DESCRIBING THE INEVITABLE POLLUTION OF NATURAL AND ORGANIC PLANTS BY GE CROPS

In Washington D.C., Congressman Dennis J. Kucinich introduced the Genetically Engineered Food Right to Know Bill in Congress at the end of 1999 requiring mandatory labeling of all GE foods and long-terms tests. In the introduction to *Imagine a World Without Monarch Butterflies*, Rep. Kucinich noted, “As the public becomes better informed of the effects and risks of genetic alteration of food, an enlightened public can pursue choices which will truly be in their best interest. . . . Government has a moral responsibility to ensure the purity and safety of the food supply. We cannot abdicate this responsibility to global corporations whose goals may be limited to profit-orientation.” To date, 58 Congresspersons support this bill, representing about 15% of the House of Representatives, and comparable legislation has been introduced in the U.S. Senate by Senator Barbara Boxer of California and Senator Daniel Patrick Moynihan of New York.

Scores of leading medical, health, and environmental organizations have come out in opposition to new GE crops and called for a moratorium on their introduction pending comprehensive scientific and medical studies. These include the British Medical Association, the Sierra Club, Friends of the Earth, Consumer’s Union, Institute of Science in Society, Alliance for Bio-Integrity, Center for Ethics and Toxics, Chefs Collaborative 2000, EarthSave, Physicians Committee for Responsible Medicine, Third World Network, and Union of Concerned Scientists.

Major food companies in the United States that have publicly pledged to stop using GE foods include Seagram Distillery (makers of corn-based whiskey), Frito-Lay (makers of corn and potato chips), Gerbers and Heinz (baby food manufacturers), MacDonald’s (the nation’s #1 user of potatoes for french fries), Iam’s (pet food supplier), Whole Foods Market, Wild Oats, Genuardi’s, and Honda Trading Co. (exporters of soybeans to Japan).

In autumn 2000, StarLink corn, a GE variety approved for animal use by the FDA but not approved for humans because of possible allergic effects, appeared in tacos and other corn products in American supermarkets and restaurants. Altogether, 300 different types of foods containing StarLink corn were contaminated, and Taco Bell, Safeway, Kellogg’s, and other suppliers undertook a massive recall. Japan, the U.S.’s largest trade partner, demanded that all corn imports from America be tested. The incident has become a nightmare for the corn industry and thousands of corn farmers in the United States. Nine million bushels of contaminated corn entered the human food supply, and Aventis, the biotech seed supplier, promised to buy back all remaining StarLink corn from farmers and manufacturers at an estimated cost of \$1 billion. The StarLink fiasco will give further impetus to

the campaign to label GE foods and products. In a scathing editorial, *Business Week* concluded: “None of this has led the biotech-food industry to soften its opposition to labeling or to any special regulations for biotech products. But it’s time for a change. Biotech foods are new, they are different, and they deserve special regulations. The industry should drop its opposition to tougher regulations.”

Benefits of Brown Rice

“At ten-thirty we were called to the ashram porch for lunch with Gandhi and the [disciples]. Today the menu included brown rice, a new selection of vegetables, and cardamom seeds.”

—PARAMHANSA YOGANANADA, AUTOBIOGRAPHY OF A YOGI

Most of the GE rice under development is designed to be milled, producing 100% polished white rice. GE nutrients or organisms are added to the starchy endosperm, so that the outer layers of germ and bran can be stripped after harvesting. Much of the plant’s protein, vitamins, and minerals are concentrated in these layers.

Since the Nobel Prize in medicine was awarded in 1929 to Christiaan Eijkman for showing that brown rice could prevent beriberi, the benefits of eating brown rice and other whole, unpolished grains have been widely recognized. Brown rice was the original staple crop of the natural foods movement in the United States in the early 1960s. Macrobiotic educators Michio and Aveline Kushi helped convince rice farmers to make available brown rice. From this tiny seed, the modern organic foods movement was born.

In the generation since, brown rice has become a staple health food in North America, Europe, Japan, and other parts of the world. Its central importance has recently been recognized in many ways, including:

- **The Food Guide Pyramid:** The latest edition of *Dietary Guidelines for Americans, 2000 version*, recognizes whole grains as the most important category of foods: “Eat foods made from a variety of whole grains—such as whole wheat, brown rice, oats, and whole grain corn—every day.” The U.S. government’s guidelines call for whole grains, vegetables, and fruits to be the center of each meal. Brown rice is mentioned throughout the text as the preferred form in which to eat rice.
- **Brown Rice for a Healthy Heart:** The American Heart Association has long recognized the importance of whole grain rice and other high fiber foods to cardiovascular health. The AHA cookbook includes recipes for brown rice. Harvard scientists reported in the early 1970s that

the macrobiotic population in Boston, eating a diet centered on brown rice, had ideal cholesterol levels and blood pressure and was virtually at no risk for heart disease. In his classic experiment showing that coronary heart disease could be reversed, as well as prevented, Dr. Dean Ornish gave his advanced heart patients a diet centered on brown rice and other whole grains.

- **Brown Rice Protects Against Cancer:** Women eating brown rice as part of a balanced macrobiotic diet process estrogen more quickly and efficiently than women eating the standard modern diet. Researchers at New England Medical Center concluded this could help prevent breast cancer. In Italy, the National Tumor Institute in Milan reported that a macrobiotic diet high in brown rice could help reduce the onset of breast cancer in high-risk women by 25 to 30 percent. Scientists at Tulane University found that patients with pancreatic and prostate cancer who observed a macrobiotic diet high in brown rice and other whole grains lived from two to three times longer than usual cancer patients. The National Cancer Institute is presently funding a clinical trial of a plant-centered diet for pancreatic cancer in which brown rice and other whole grains are a major component of the diet.

- **Whole Rice Reduces Risk of Diabetes:** White rice and other starchy foods contribute to diabetes, according to Harvard University researchers. A 1997 study of 65,173 women who ate a high starch diet that was low in fiber had 2.5 times as much diabetes as women who consumed whole grains. "Bread, rice, and pasta should be in the whole grain form; brown rice and whole-grain pastas and breads," explained Dr. Walter Willet, professor of epidemiology and nutrition at the Harvard School of Public Health.

- **Schools Offer Healthy Lunches:** Across the country, school districts are beginning to offer brown rice, stir-fired vegetables, salad bars, and other healthy foods. In a survey, the Physicians Committee for Responsible Medicine reported that public schools in Miami, New York City, Houston, and Atlanta offered daily vegetarian or low-fat lunch entrées.

- **Whole Foods Behind Bars:** The U.S. Bureau of Prisons began offering federal prisoners a choice of plant-based foods at every meal on Oct. 1, 2000. "The intent of the meatless alternatives is to meet the changing dietary habits and the religious dietary needs of the diverse inmate population," a spokesperson noted. Whole grains, soyburgers, barbecued tofu, fried eggplant, and other foods will be made available at 98 prisons housing about 126,000 inmates.

- **Brown Rice in Outer Space:** NASA is developing a special all-plant diet for astronauts featuring brown rice, lentils, seitan, whole wheat tortillas, garden vegetables, and desserts made with amasake, a brown rice-based sweetener.
- **Asian Scientists Promote Brown Rice:** In Los Banos, Philippines, the “Mecca for the world’s rice eaters,” former scientists for the International Rice Research Institute (IRRI), the center of world rice biotechnology, have set out to convert the nation from eating white rice to brown rice. In autumn 2000, 75 scientists attended the first meeting of Los Banos Pinawa (*pinawa* is the Tagalog word for brown rice). “If we can get the elite to start eating brown rice, we think the masses will imitate,” Dr. Ben Vergara, a retired IRRI plant physiologist, told the *Asian Wall Street Journal*. He said that brown rice contained 12% more protein, 100% more fiber, essential minerals and vitamin B, and far more of the anti-cancer compound, phytate, than white rice. “There are lots of new dishes being invested to make use of our wonderful brown jasmine rice,” exclaimed Dr. Kwanchai Gomez.
- **King of Siam Eats Brown Rice:** The Thai government led a campaign to promote brown rice in 1999 for health and environmental reasons. The national Electricity Generating Authority spearheaded an effort to reduce energy by 60% in the country’s 30,000 rice mills, most of which was going to produce white rice. King Bhumiphol, the nation’s monarch, endorsed the campaign with a public statement that he eats brown rice. Bangkok supermarkets, restaurants, and Thai airlines now regularly offer brown rice.

Natural Abundance

“A grain of wheat will produce ten thousand heads, and every head will have ten thousand grains, and every grain will produce ten pounds of fine clean flour. And other seeds, fruits, and grass will produce in corresponding proportions.”—Jesus quoted by Irenaeus

“From one grain, ten thousand grains.”—George Ohsawa

“In a quiet village on the outskirts of Zurich, a genetically engineered strain of rice . . . in locked up in a grenade-proof greenhouse as if it were the Frankenstein monster that some critics contend it is.”—“Golden Rice in a Grenade-Proof Greenhouse.”—NEW YORK TIMES, NOVEMBER 21, 2000

The world's traditional rice harvest is in imminent peril. In America genetically engineered corn displaced one third of the natural corn crop in just a few short years. Now organic and conventional corn is in danger of being polluted from surrounding GE corn. The Monarch butterfly population is at risk, and other adverse health and environmental effects may emerge in the future. The same pattern could happen when GE rice is introduced into America and other countries this year or next.

The biotech agriculture industry has launched a \$50 million public relations campaign to promote GE foods and oppose those who question these new technologies. In touting the benefits of GE rice, it claims that brown rice and other whole grains are poor nutritional sources of certain vitamins and minerals. The spokesman for biotechnology is Dr. Ingo Potrykus, a professor of plant biology at the Institute of Plant Sciences in Zurich and a developer of GE rice. He appeared on the cover of *Time* magazine on July 4, 2000 for a special report on how new "golden rice" will save the lives of millions of children. The *Time* article was highly selective, and among the things it left out were recent remarks of Dr. Potrykus to an international congress of 4000 scientists in which he said that natural rice is "notoriously deficient" in iron and vitamin A. The biotech industry also recycles outdated and misleading claims that the phytic acid found in natural rice interferes with iron absorption, a view that has been contradicted by more recent research as noted above. In subsequent remarks, Dr. Potrykus, who protects his artificial creation in a bomb-proof bunker, likened opposition to genetic engineering as "well-poisoning" and characterized those who question GE rice as "criminal." Presumably this included Germany's agriculture minister, Karl-Heinz Funke, who attended the same symposium as Potrykus and defended Europe's opposition to widespread use of GE crops. "People in Germany associate biotechnology and genetic engineering with moral deficits because too many questions remain unanswered," Funke said. "Add to this, historical reasons which might play a role for us Germans. This sows the seeds of mistrust. . . . A good many people can't help suspecting that this is all about commercial gain."

The aim of the biotech campaign is not only to convince the public that GE foods are superior but also that natural foods are inferior, nutritionally deficient, and poorly absorbed. Biotech scientists routinely contrast low levels and poor absorption of iron in diets of vegetable origin (e.g., "unrefined cereals including rice, nuts, and dark leafy vegetables") with animal food diets "containing blood and muscle." On 20-20, ABC-TV's news magazine, biotech exponent Dennis Avery recently attacked organic food, charging that organic fertilizers were contributing to the *E coli* epidemic and labeled claims that organic foods are more nutritious as fraudulent. The *Sacramento Bee*, the principal newspaper in the California rice-growing region, has rushed to embrace "golden rice" and other GE varieties. "The re-

searchers' golden rice is not a Frankenfood," the paper editorialized, demonizing those who question its value. "It is those who seek to keep this rice locked away, because it doesn't fit their ideology about the evils of modern science, who are acting like monsters."

The time to act is now. Rice and other whole grains that have nourished and sustained humanity's biological and spiritual evolution are at risk. An informed citizenry and enlightened farmers, manufacturers, cooks, scientists, physicians, distributors, suppliers, and others must join together and preserve organic rice and natural strains that have been cultivated for millennia. The California Rice Industry is clearly more attuned to the health and environmental risks than other sectors of modern agriculture. It is proceeding cautiously to embrace GE rice, but appears likely to back off if consumers mobilize against it. Ultimately, the marketplace will decide whether GE rice takes root in California and other regions of America. As Greenpeace concluded in its report on farming in the state, "California is clearly at a crossroads. It can choose genetic engineering or it can choose organic agriculture. Both cannot coexist in the state."

Those of us who have been eating organic rice and using whole foods to heal, maintain our health, and realize our dreams now have a golden—amber—opportunity to show our gratitude by joining together in this sacred task. We invite you to become actively involved with Amberwaves, a network devoted to preserving natural and organic rice and other whole grains for future generations. In the beginning, our power to effect change will be very small, but like rice itself, over time it will grow and multiply. From one grain, ten thousand grains will bloom, creating enduring beauty, health, and peace.

2

Consumer's Guide to GE Rice and Other Grains

By Alex Jack

"Genetically modified means an organism in which the genetic material has been changed through gene technology in a way that does not occur naturally by multiplication and/or any natural recombination."

—United Nations, 2000

Scores of genetically modified foods and products have been introduced around the world. For a comprehensive list of products that have been approved in the United States, please see *Imagine a World Without Monarch Butterflies* by Alex Jack (One Peaceful World Press, 2000). This section deals primarily with GE grains in development or grain products currently on the market that contain GE ingredients.

Countries Producing GE Foods

At the present time, the following countries are growing or producing GE foods and products:

- United States (50+ foods in total)
- Canada (30 foods)
- Japan (22 varieties of 6 crops)
- European Union, especially Spain and France (9 foods)
- Argentina (3 foods)
- Mexico (3 foods)
- Australia (2 crops + 6 field trials)
- Brazil (1 food)
- South Africa (1 food)
- China (1 crop)
- Russia (1 crop)
- India (1 crop)
- Vietnam (1 crop)

Countries Developing GE Rice

- United States
- China
- Japan
- Korea
- Philippines
- Switzerland
- Mexico
- Australia
- Italy

Countries Limiting GE Foods

Countries with labeling laws, restrictive agriculture, environmental, or import policies, or prohibitions:

- | | |
|--------------------------|-------------------------|
| • Australia | • Namibia (banned) |
| • Austria | • New Zealand |
| • Brazil (attempted ban) | • Norway (banned) |
| • Denmark | • Russia |
| • Ethiopia | • Saudi Arabia (banned) |
| • France | • Sri Lanka (banned) |
| • Germany | • Switzerland |
| • Greece | • Taiwan |
| • Italy | • Tasmania (banned) |
| • Luxembourg | • Thailand |
| • Japan | • Uganda (banned) |
| • Korea | • United Kingdom |
| • Mexico | • Wales (banned) |

Major Biotech Companies

- Aventis (French company combining Hoechst and Rhone Poulenc)
- Monsanto (U.S. company, subsidiary of Pharmacia and Upjohn)
- DuPont (U.S. company operating under Pioneer Hybrid division)
- Syngenta (Pending merger of Novartis and AstraZeneca in Europe)
- Dow Chemical (U.S. company)

Rice Species

- Indica (*Oryza sativa*, grown in South Asia, Southeast Asia, Southern United States)
- Japonica (*Oryza sativa*, grown in the Far East, California)
- Africana (*Oryza glaberrima*, grown in Africa)
- Javonica (*Oryza sativa*, grown in Indonesia)

Rice Varieties

- Short grain
- Medium grain
- Long grain
- Sweet rice
- Basmati rice, jasmine rice, etc.

Rice Products

Rice is hulled, milled, or processed in a variety of ways. The main types of rice and rice products available include:

- Whole grain rice (brown rice, unpolished rice, unhulled rice)
- White rice (polished rice, converted rice, parboiled rice)
- Amasake (sweet rice beverage)
- Rice cakes
- Rice syrup
- Rice milk
- Mochi (pounded sweet rice)
- Rice miso (soybean paste with rice)
- Rice noodles
- Rice balls
- Sushi
- Rice crackers
- Puffed rice
- Rice dumplings
- Rice croquettes
- Rice cream
- Rice porridge
- Rice pilaf
- Rice bread
- Rice flour
- Rice flakes
- Rice bran
- Rice cookies
- Rice salad
- Rice pudding
- Rice custard
- Rice vinegar
- Rice bran oil
- Rice tea
- Beer
- Sake (rice wine)

GE Rice Research and Development

Several dozen varieties of GE rice are under development or being field tested. Between 1982 and 1997, 160 patents were granted or pending. In 2001, the mapping of the rice genome was completed, spurring further GE development. Major GE rice developers and the traits they are modifying include:

- AgrEvo (male sterility)
- American Cyanamid (herbicide resistance)
- AstraZeneca (beta-carotene content)
- Aventis (herbicide resistance)
- Catholic University of Piacenza (insect resistance)
- Chinese government (insect resistance)
- Chonnam National University, Korea (higher yields)
- Du Pont (protein content)
- Institute of Genetics, Berlin (starch content)
- Institute of Plant Sciences, Zurich (beta-carotene content)
- International Rice Research Institute (beta-carotene content)
- Japan Tobacco (altered photosynthesis/flowering)
- Kubota (yield increase, protein content)
- Louisiana State University (bacterial resistance)
- Misubishi (viral resistance)
- Mitsui (yield increase, starch content)
- Monsanto (herbicide tolerance)
- Novartis (herbicide tolerance)
- Pioneer Hi-Bred (pest resistance and fungal resistance)
- Rockefeller Foundation (enhanced nutrition)
- Science University of Tokyo (hepatitis vaccine)
- University of California (bacterial resistance)
- Washington State University (bigger grains)
- West Africa Rice Development Association (higher yields)

GE Rice and Rice Products

The first GE rice designed for commercial production was approved by the U.S.D.A. in December 2000. GE varieties under development include:

- Cadet and Jacinto (U.S.D.A., Agricultural Research Service, Beaumont, Texas, improved cooked rice texture)
- IMI Rice Seed (American Cyanamid, herbicide-resistance)
- Liberty Link Rice (AgrEvo and Louisiana State University, herbicide-resistance)
- Roundup-Ready Rice (Monsanto, herbicide-resistance)

GE Wheat and Wheat Products

In early 2001, Monsanto launched the first field trials of Roundup Ready spring wheat in North Dakota, South Dakota, Minnesota, and Montana. Like its altered corn and soybeans, GE wheat is designed to be resistant to the herbicide, Roundup. Scientists at Oregon State University, Washington State University, University of South Dakota, and University of Minnesota are engaged in the effort. Commercial production would begin in 2003, and red winter wheat and other varieties would follow.

U.S. Wheat Associates, the major trade organization, has declined to take any action in support of GE wheat. But it has called for systems to prevent the commingling of GE wheat with conventional varieties if it is introduced. On general issues related to biotechnology, the Wheat Associates' board endorsed voluntary food labeling of GE ingredients and supported the consumer's right to know and the food industry's right to inform.

In Canada, which is a major wheat producer, Canadian Wheat Board CEO Greg Arason stated, "No transgenic varieties should be registered for commercial production in Canada until either they have achieved full commercial acceptance in all their potential markets, or until we have cost effective technologies to segregate by variety throughout the system." Arason called for the freeze on the registration of new transgenic crops at a meeting of the Canada Grains Council.

Although GE wheat is not available, it is being grown on test plots, and in 1999 Thailand reported that a shipment of wheat from the Pacific Northwest was contaminated with experimental GE wheat.

Many common wheat products, however, may contain other GE ingredients, especially altered canola oil, cottonseed oil, or peanut oil. Yeasted bread, especially that made with white flour, may contain GE enzymes.

GE Enzymes Used in Wheat Products

- Alpha Amylase (cereals, beer, and other foods)
- Hemicellulase (bread flour)
- Lipase and Triacylglycerol (baked goods)
- Novamyl (bread and baked goods)
- Pullulanase (high fructose corn syrup)

Wheat Products That May Contain GE Enzymes

- | | |
|------------|-------------|
| • Bread | • Doughnuts |
| • Crackers | • Cakes |
| • Cookies | • Pies |
| • Bagels | • Pastries |

GE Corn and Corn Products

Corn is the only genetically engineered grain currently on the world market. An estimated 25% of the U.S. corn crop, including corn for both animal feed and human consumption, was modified in 2000. (This is down from a high of 35% in 1999.) The major type of GE corn contains Bt, a bacterium that releases a toxic protein that is designed to kill the corn borer and other pests. However, the plant's pollen can endanger beneficial insects, birds, and other animals as well. In experiments, Cornell University scientists reported that pollen from GE corn that drifted to adjacent milkweed plants killed 44% of the larvae of Monarch butterflies within several days. Many processed food products in the supermarket and natural foods store contain corn syrup, cornstarch, corn dextrose, corn oil, corn flour, or other corn product that may be genetically altered. Ninety percent of the corn crop is fed to livestock to produce animal foods.

GE Corn and Corn Products

- Fresh corn
- Popcorn
- Corn tortillas
- Grits
- Polenta
- Masa (corn dough)
- Corn syrup
- Corn fructose
- Corn starch
- Corn dextrose
- Corn oil
- Corn flour
- Cornmeal
- Other corn products

Processed Foods and Products That May Contain GE Corn

- Corn Chips
- Cookies, Candies, and Gum
- Bread
- Cereals
- Pickles
- Margarine
- Beer and Alcohol
- Soft Drinks, Spritzers, Fruit Drinks
- Enriched Flours and Pastas
- Salad Dressings
- Vanilla
- Infant Formula
- Vitamin C tablets

Products Produced from Animals Fed GE Corn and Soy

- Beef, including hamburger, steak, etc.
- Pork, ham, hot dogs
- Lamb
- Chicken, eggs, turkey, and other poultry
- Factory-fed trout, salmon, and other fish

GE Barley and Barley Products

Australia, the United Kingdom, and several other countries are experimenting with GE barley. However, none is commercially available yet. The malting barley industry, which manufactures beer, reportedly will not accept GE malting barley in any country except China.

Selected Companies Using GE Ingredients in Grain Products

- Aunt Jemima (pancake and waffle mixes)
- Beechnut (baby foods)
- Betty Crocker (pie and pancake mixes)
- Bisquick (mixes)
- Campbell's (corn chowder, chicken rice, chicken noodle soup)
- Celeste (pizza)
- Delicious (cookies)
- Duncan Hines (mixes)
- Frookies (cookies)
- General Mills (Cheerios, Wheaties, breakfast cereals)
- Green Giant (harvest burger)
- Keebler (cookies)
- Kellogg's (corn flakes, fruit and grain bars)
- Kraft (tacos, corn products)
- Morningstar (vegetarian burgers)
- Nabisco (fruit and grain bars, Oreo cookies)
- Nature Valley (granola bars)
- Old El Paso (taco shells)
- Pepperidge Farms (breads, baked goods, corn chowder)
- Pillsbury (bread and muffin mixes)
- Post (breakfast cereals)
- Quaker Oats (breakfast cereals)
- Sesame Street (cookies)
- Thomas's (English muffins)
- Tombstone (pizza)
- Wonder (bread)

Selected Companies Not Using GE Ingredients in Grain Products

- A. C. LaRocco (pizza)
- Amy's Kitchen (pizza)
- Arrowhead Mills (breakfast cereals)
- Barbara's Bakeries (breakfast cereals, cookies)
- Bearitos (corn chips, natural foods)
- Earth First (breakfast cereals, natural foods)
- Earth's Best (baby foods)
- Eden Foods (rice and soymilk, pastas, and other products)
- EnviroKidz (natural foods)
- Erewhon (rice crispies and other products)
- French Meadows (bread)
- Frito-Lay (corn chips)
- Garden of Eatin' (blue corn chips)
- Gelentano (frozen vegetable lasagna)
- Genuardi's (brand name supermarket products)

- Gerber's (baby foods)
- Health Valley (granola bars, cookies)
- Heinz (baby foods)
- Kirin and Sapporo (corn-based beer imported from Japan)
- Lifestream (breakfast cereals, natural foods)
- Lundberg Farms (brown rice, rice cakes, rice syrup)
- Manna (bread)
- McDonald's, Burger King, and Procter & Gamble (just french fries)
- Millina's Finest (pasta)
- Nature's Path (cereals)
- New Pioneer Co-op (bread)
- Novartis (Ovaltine, Gerbers, Wasa crackers, health food lines)
- Pamela's (cookies)
- Seagram (corn-based liquor)
- Shiloh Farm (bread)
- Turtle Island (grain tempeh)
- Vita Spelt (spelt)
- White Wave (grain tempeh)
- Whole Foods (brand name natural supermarket products)
- Wild Oats (brand name natural supermarket products)

GE-Free Seeds

Fifty seed companies around the country have joined in a Safe Seed Initiative, pledging "not to knowingly buy or sell GE seeds or plants." These include:

- Abundant Life Seed Foundation
- Alberta Nurseries & Seeds
- Heirloom Seed Project
- Heirloom Seeds
- Johnny's Selected Seeds
- KUSA Seed Research Foundation
- New England Seed Co.
- Prairie Garden Seeds
- Seed Savers Exchange
- Seeds for the South
- Seeds of Change
- Thomas Jefferson Center for Historic Plants

3

From 10,000 Grains, One Grain

By Edward Esko



Edward Esko (right) with Grant Lundberg, CEO of Lundberg Family Farms, leading producer of organic brown rice in the United States.

"For every new engineered seed that makes it to field trials, 10,000 failed during the development."

— PEGGY G. LEMAUX, A SCIENTIST AT THE UNIVERSITY OF CALIFORNIA, DAVIS, FROM AN ADDRESS TO THE ANNUAL RICE GROWERS MEETING IN CALIFORNIA

Edward Esko has lectured on diet, holistic health, and planetary ecology around the world. He is the author of Healing Planet Earth, Yin Yang Primer, and Contemporary Macrobiotics.

The Visitor's Center at Lake Oroville sits high above the Sacramento Valley in the foothills of the Sierras. Like most days in the Valley, the weather is clear, sunny, and warm. Perfect for growing rice. The Sacramento Valley, which stretches off toward the West, is the leading producer of organic brown rice in the United States. Alex Jack (author of *Let Food Be Thy Medicine*) and I stand on the upper deck of the observation tower and trace the pattern of rivers flowing down from the Sierras. The panorama is breathtaking. Streams and rivers flow steadily downward from the high mountains. Their energy gathers several hundred feet below us in Lake Oroville.

Lake Oroville was created following construction of Oroville Dam in 1968. Made of earth, the dam rises seven hundred feet above the bed of the Feather River. Rice cultivation requires a steady supply of water. Lake Oroville is the source of water for the rice fields in the Sacramento Valley.

Sandwiched between the Pacific Ocean and the Pacific Coastal Range to the West, and the Sierra Nevadas to the East, the Sacramento Valley is charged by natural energy. To the North lie Mount Shasta and the Cascades.

Mountain ranges are the earth's energy meridians. They are formed by the upward push of energy generated by the earth's rotation. Rivers are ley lines that channel energy from the mountains. Farmers in the Sacramento Valley further channel this energy through a vast network of canals that crisscross the Valley and irrigate the rice fields. The successful channeling of this mountain energy has made California an agricultural powerhouse, both for conventional and organic farming.

The reason for my visit to California is to meet with organic rice farmers to discuss the status of genetically altered rice. Alex Jack and I are planning to meet with Grant Lundberg, the C.E.O. of Lundberg Family Farms. The Lundberg family has been growing rice since 1937. Today, Lundberg Farms is the largest distributor of organic brown rice in the United States.

There are currently a dozen varieties of genetically engineered (GE) rice in development around the world. If GE rice comes into the Sacramento Valley, there is a danger of genetic pollution caused by GE pollen drifting into the organic fields and contaminating the organic crop. America could lose its primary source of organic brown rice. Heirloom rice that has nourished humanity for thousands of years could conceivably vanish. Like wheat, barley, and the other cereal grains, rice is at the center of the human evolutionary spiral. Fifty-one percent of the world's population consumes rice as daily staple. Grains are the cornerstone of our agriculture, diet, and civilization. The genetic engineering of such an essential food could have negative implications for our entire planet.

On the following day, we set out for our meeting. On the drive from Oroville to Richvale, the home of Lundberg Farms, there are rice fields as

far as the eye can see. Off in the distance are metal silos used for storing the rice crop. It is early summer, and the rice resembles tall green grass. Surrounding the fields are irrigation ditches in which cattails grow. Dragonflies swarm above. The ecology is that of a freshwater marsh.

Upon arriving at Lundberg Farms, we locate Grant Lundberg's office in a small shed next to the main office building. After shaking hands, Grant leads us into his office. We begin by introducing ourselves and thanking the Lundbergs for their pioneering efforts in developing and promoting organic rice. Alex then presents a scenario in which a dozen varieties of GE rice are introduced around the world. Grant is well aware of the plans for GE rice. He obviously shares our concerns. We then ask how California farmers plan to deal with this threat.

Grant explains that farmers in California are wary of GEOs (genetically engineered organisms.) Some, like the Lundbergs, are opposed to GEOs for philosophical reasons. Others are concerned about their economic implication. Rice farmers know that the vast majority of American consumers favor labeling of GEOs. Moreover, forty percent of the California rice crop is exported, mostly to Japan. The Japanese recently passed a law requiring that all GE products be labeled.

These concerns prompted California farmers to work with state legislators in Sacramento to pass a "seed certification law." The law requires that GE crops be strictly segregated from conventional and organic crops. Grant expresses his hope that the seed certification law will at the very least protect non-GE organic and conventional crops from genetic contamination.

The biotech industry has ambitious plans for California rice. Their hope is to see 40 percent of the California crop be GE by 2010. The first two varieties of GE rice set for introduction in California are the herbicide tolerant varieties "Liberty Link," developed by AgrEvo, and "Roundup Ready," developed by Monsanto. Herbicide tolerant rice is genetically engineered to accept high levels of "Liberty," the herbicide marketed by AgrEvo, and "Roundup," the herbicide marketed by Monsanto.

Grant then contrasts the Lundberg method of organic weed control with the use of Roundup and other toxic herbicides. During the summer, the Lundbergs drain the water from their organic fields. The bright sun shines down on both weeds and rice plants. The weeds dry up but the rice flourishes. The fields are again flooded and the rice continues growing without competing weeds—a totally natural method of weed control.

Several days later, after returning to Massachusetts, I notice the latest edition of *Time* magazine. The cover features a photo of a Swiss Professor named Ingo Potrykus with his beta-carotene-enriched rice. The headline states, "This Rice Could Save a Million Kids a Year. . . . but protesters believe such genetically modified foods are bad for us and our planet." So-

called “golden rice,” an unnatural transgenic “food,” contains beta carotene producing genes from daffodils and a certain type of bacteria. For many years, Potrykus has been waging a one-man campaign against brown rice; claiming it is nutritionally deficient. However, beta carotene is present in the outer coat of brown rice; the part that is removed when it is milled into white rice. Most of the nutritional deficiencies in underdeveloped countries occur because people eat white, rather than brown rice, and not enough green leafy and orange-yellow vegetables.

Scientists such as Potrykus begin from a false premise. They believe nature is imperfect and can be improved upon. Such blindness inevitably leads to downfall. Nature is and always will be a perfect system. However, in the case of “golden rice,” this blindness is being presented under a humanitarian guise. I hope people are not fooled.

Rice farming in the Sacramento Valley has been going on for over a hundred years. GEOs are simply not needed; they interfere with the natural order and make things unnecessarily complicated. They are more of a problem than a solution. There are far better solutions to hunger and malnutrition than genetic engineering. GEOs divert attention from the root cause of these problems and make finding solutions more difficult. During our meeting, Grant, Alex, and I agreed that things would be much better if GEOs simply went away.

However, the reality is that they are not going to go away, at least not in the near future. Therefore, we must remain vigilant; we must make our choices and our voices heard. Together we can send a strong message that we wish to see nature respected on all levels.

Nature operates according to the principle of “one grain, ten-thousand grains.” For every seed the earth receives, it returns thousands. Nature is forever productive, and infinitely diverse. In the practice of genetic engineering, ten-thousand seeds are sacrificed for every one that comes to market, or “from ten-thousand grains, come one grain.” Originally, the heirloom seeds that sustained humanity over the centuries cost nothing; they were given freely by nature. Each genetically engineered seed costs \$300 million to bring to market. We now face a choice between two opposite views of life, one natural, the other artificial. The health and happiness of future generations, indeed the health and happiness of all species, could well depend upon which of these two paths we choose.

4

GE Vitamin 'A' Rice: A Blind Approach to Blindness Prevention

By Dr. Vandana Shiva

Genetically engineered vitamin A rice has been proclaimed as a miracle cure for blindness —“a breakthrough in efforts to improve the health of billions of poor people, most of them in Asia.”

More than \$100 million have been spent over 10 years to produce a transgenic rice at the Institute of Plant Sciences at the Swiss Federal Institute of Technology in Zurich. The Zurich research team headed by Ingo Potrykens and Xudong Ye introduced three genes taken from a daffodil and a bacterium into a rice strain to produce a yellow rice with high levels of beta-carotene, which is converted to vitamin A within the body.

The rice is being promoted as a cure for blindness since vitamin A deficiency causes vision impairment and can lead to blindness. According to the UN, more than 2 million children are at risk due to vitamin A deficiency.

The work in Zurich was funded by grants from the Rockefeller Foundation, the agency which had launched the chemical agriculture in Asia through the Green Revolution which led to erosion of biodiversity and erosion of diverse sources of nutrition for the poor. In addition, the Swiss Government and the European Community has also supported the research.

Vandana Shiva is a physicist and environmentalist from India. A leader of the international movement against GE foods, she is founder of the Third World Network, director of the Research Foundation for Science, Technology, and Natural Resource Policy, and recipient of the Alternative Nobel Peace Prize. Her books include Stolen Harvest, Biopiracy: The Plunder of Nature and Knowledge, and The Violence of the Green Revolution. © 2000 by Vandana Shiva.

Vitamin A Rice Is a Hoax

The problem is that vitamin A rice will not remove vitamin A deficiency (VAD). It will seriously aggravate it. It is a technology that fails in its promise.

Currently, it is not even known how much vitamin A the genetically engineered rice will produce. The goal is 33.3% micrograms/100g of rice. Even if this goal is reached after a few years, it will be totally ineffective in removing VAD.

Since the daily average requirement of vitamin A is 750 micrograms of vitamin A and 1 serving contains 30g of rice according to dry weight basis, vitamin A rice would only provide 9.9 micrograms which is 1.32% of the required allowance. Even taking the 100g figure of daily consumption of rice used in the technology transfer paper would only provide 4.4% of the RDA.

In order to meet the full needs of 750 micrograms of vitamin A from rice, an adult would have to consume 2 kg 272g of rice per day. This implies that one family member would consume the entire family ration of 10 kg. from the PDS in 4 days to meet vitamin A needs through "Golden rice."

This is a recipe for creating hunger and malnutrition, not solving it.

It will however take millions more in dollars and another decade of development work at the International Rice Research Institute to produce vitamin A rice varieties that can be grown in farmers fields.

Is the "golden" rice a miracle that is the only means for preventing blindness for Asia or will it introduce new ecological problems like the Green Revolution did and create new health hazards like other genetically engineered foods?

The genetic engineering of vitamin A rice deepens the genetic reductionism of the Green Revolution. Instead of millions of farmers breeding and growing thousands of crop varieties to adapt to diverse ecosystems and diverse food systems, the Green Revolution reduced agriculture to a few varieties of a few crops (mainly rice, wheat, and maize) bred in one centralized research centre (IRRI for rice and CIMMYT for wheat and maize).

The Green Revolution led to massive genetic erosion in farmers fields and knowledge, erosion among farming communities, besides leading to large scale environmental pollution due to use of toxic agrichemicals and wasteful use of water.

Genetically engineered rice as part of the second Green Revolution is repeating the mistakes of the Green Revolution while adding new hazards in terms of ecological and health risks.

The "selling" of vitamin A rice as a miracle cure for blindness is based on

blindness to alternatives for removing vitamin A deficiency and blindness to the unknown risks of producing Vitamin A through genetic engineering.

Eclipsing Alternatives

The first deficiency of genetic engineering rice to produce vitamin A is the eclipsing of alternative sources of vitamin A. Per Pinstripe Anderson, head of the International Rice Research Institute, has said that vitamin A rice is necessary for the poor in Asia, because “we cannot reach very many of the malnourished in the world with pills.” However, there are many alternatives to pills for vitamin A supply. Vitamin A is provided by liver, egg yolk, chicken, meat, milk, butter. Beta-carotene, the vitamin A precursor is provided by dark green leafy vegetables, spinach, carrot, pumpkin, mango, and drumstick. Women farmers in Bengal use more than 100 plants for green leafy vegetables.

The lower cost, more accessible and safer alternative to genetically engineered rice is to increase biodiversity in agriculture. Further, since those who suffer from vitamin A deficiency suffer from malnutrition generally, increasing the food security and nutritional security of the poor through increasing the diversity of crops and diversity of diets of poor people who suffer the highest rates of deficiency is the reliable means for overcoming nutritional deficiencies.

Sources of vitamin A in the form of green leafy vegetables are being destroyed by the Green Revolution and genetic engineering which promote the use of herbicides in agriculture. The spread of herbicide resistant crops will further aggravate this biodiversity erosion with major consequences for increase in nutritional deficiency. For example, *bathua*, a very popular leafy vegetable in North India, has been pushed to extinction in Green Revolution areas where intensive herbicide use is a part of the chemical package.

Environmental Costs of Vitamin A Rice

Vitamin A from native greens and fruits is produced without irrigation and wastage of scarce water resources. Introducing vitamin A in rice implies a shift from water conserving alternatives for vitamin A to water intensive system of production since so called high yielding rice varieties are highly water demanding. vitamin A rice will therefore lead to mining of ground water or intensive irrigation from large dams with all the associated environmental problems of water-logging and salinisation.

Further, as in the case of other genetically engineered crops, rice with vitamin A will have impact on the food web. The ecological impact on soil organisms and other organisms dependent on rice in the food chain should be part of the biosafety analysis of genetically engineered rice before it is released for production. Research has already shown that indigenous rice va-

ieties support far more species than Green Revolution varieties. How will genetically engineered rice impact biodiversity and the potential for disease and pest vulnerability?

Health Risks of Vitamin A Rice

Since rice is a staple eaten in large quantities in Asian societies, vitamin A rice could lead to excessive intake of vitamin A especially among those who do not suffer from vitamin A deficiency. Excess vitamin A can lead to hypervitaminosis A or vitamin A toxicity. Such toxicity is known to occur due to over ingestion of vitamin A rich food e.g., polar bear liver or by food faddism by over solicitous parents, or as side effects of inappropriate therapy.

Vitamin A toxicity can lead to abdominal pain, nausea, vomiting, dizziness, popillidena, bulging fontanelle.

Chronic toxicity of vitamin A can occur after ingestion of large quantities of vitamin A for protracted periods. Chronic vitamin A toxicity is characterized by bone and joint pain, hyperotosis, hair loss, dryness and fissures of lips, nausea, intracranial hypertension, low grade fever, pruritis, weight loss, hepatosplenomegaly.

Natural sources of vitamin A are consumed seasonally and in small quantities as greens, relishes, fruits and hence do not carry the risks of vitamin A toxicity. Rice eating regions have been found to be associated with higher malnutrition than wheat eating regions, especially after the Green Revolution, which destroyed fish and plant biodiversity necessary for a balanced diet. These regions also have higher prevalence of water borne diseases like diarrhea, amoebiasis hepatitis A and E, dysentery, and vector borne diseases like malaria, which unlike in earlier years when it was a less hazardous form of malaria caused by *plasmodium vivax* is increasingly becoming falciparum malaria. These health problems are known to involve damage to the liver. The additional risks of vitamin A toxicity under these conditions of vulnerable health situation of the poor in Asia needs to be assessed with care before a large scale push is given to genetically engineered rice.

Further, the globalisation of agriculture is leading to an increase in malnutrition in the Third World, as the most fertile ecosystems are diverted to luxury export crops, and as domestic markets are destroyed due to dumping of subsidized agricultural commodities. In India, per capita consumption of cereals has declined by 12 per cent in rural areas over the past two decades. The shift from policies based on the "right to food" to free trade policies will push millions into hunger and poverty.

Genetically engineered rice is part of a package of globalized agriculture which is creating malnutrition. It cannot solve the problems of nutritional deficiency but it can introduce new risks of food safety. Since the vitamin A

in rice is not naturally occurring and is genetically engineered, novel health risks posed by vitamin A rice will need to be investigated before the rice is promoted by IRRI and aid agencies or commercialized.

The risk assessment for living modified organisms intended for direct use as feed is given in Annexe II of the recently finalized Biosafety Protocol under the Convention on Biological Diversity.

The risk assessment of vitamin A rice should therefore involve the following steps:

- An identification of any novel genotypic and phenotypic characteristics associated with the vitamin A rice that may have adverse effects on biological diversity in the likely potential receiving environment, taking also into account risks to human health.
- An evaluation of the likelihood of these adverse effects being realized, taking into account the level and kind of exposure of the likely potential receiving environment.
- An evaluation of the consequences should these adverse effects be realized.

The risk assessment also needs to take into account the vectors used, the insects, the ecological differences between transgenic vitamin A rice, and conventional rice varieties. The diverse contexts in which the rice is to be potentially introduced also needs to be taken into account. This includes information on the location, geographical, climatic and ecological characteristics, including relevant information on biological diversity and centres of origin of the likely potential receiving environment.

It is these potential risks which have put a question mark on genetic engineering in agriculture. The genetically engineered vitamin A rice is now being used as a Trojan horse to push genetically engineered crops and foods.

Mr. Pinstруп Anderson, the IRRI Director, has suggested that the “vitamin A rice could provide a public relations boost for plant biotechnology, which has been criticized by some environmentalists and consumer activists for promoting ‘Franken foods.’” It has yet to be established that genetically engineered rice is not a Franken food.

But one thing is clear. Promoting it as a tool against blindness while ignoring safer, cheaper, available alternatives provided by our rich agrobiodiversity is nothing short of a blind approach to blindness control.

5

Tale of Two Botanies

By Amory B. and L. Hunter Lovins

Plants, shaped into incredible diversity by 3.8 billion years of evolution, make possible all life and are resilient against almost any threat—except human destructiveness. From botany came the genetics of Mendel and Lamarck, formalizing the patient plant-breeding that created 10,000 years of agriculture.

Now, however, in the name of feeding a growing human population, a completely different kind of botany, in the Cartesian tradition of reducing complex wholes to simple parts, strives to alter isolated genes while disregarding the interactive totality of ecosystems. Its ambition is to replace nature's wisdom with people's cleverness; to treat nature not as model and mentor but as a set of limits to be evaded when inconvenient; not to study nature but to restructure it.

The new botany aligns the development of plants with their economic, not evolutionary, success: survival not of the fittest but of the fattest. High-yield, open-pollinated seeds abound; the new crops were created not because they're productive but because they're patentable. Their economic value is oriented not toward helping subsistence farmers to feed themselves but toward feeding more livestock for the already overfed rich. Most worryingly, the transformation of plant genetics is being accelerated from the measured pace of biological evolution to the speed of next quarter's earnings report. Such haste makes it impossible to foresee and forestall: Unintended consequences appear only later, when they may not be fixable, because novel life forms aren't recallable.

Amory B. Lovins, a MacArthur Fellow and consultant physicist, and L. Hunter Lovins, a lawyer and social scientist, are consultants to industry; co-founders and co-CEOs of the Rocky Mountain Institute; and co-authors with Paul Hawken of Natural Capitalism: Creating the Next Industrial Revolution.. A condensation of this article first appeared in the August 1, 1999 St. Louis Post-Dispatch under the title "Replacing Nature's Wisdom with Human Cleverness." © 1999 Rocky Mountain Institute.

In nature, all experiments are rigorously tested over eons. Single mutations venture into an unforgiving ecosystem and test their mettle. What's alive today is what worked; only successes yield progeny. But in the brave new world of artifice, organisms are briefly tested by their creators in laboratory and field, then mass-marketed worldwide. The USDA has already approved about 50 genetically engineered crops for unlimited release; U.S. researchers have tested about 4,500 more.

Over half the world's soybeans and a third of the corn now contain genes spliced in from other forms of life. You've probably eaten some lately—unwittingly. The official assumption is that they're different enough to patent but similar enough to make identical food; Europe's insistence on labeling, to let people choose what they're eating, is considered an irrational barrier to free trade.

Traditional agronomy transfers genes between plants whose kinship lets them interbreed. The new botany mechanically transfers genes between organisms that can never mate naturally: An antifreeze gene from a fish becomes part of a strawberry. Such patchwork, done by people who've seldom studied evolutionary biology and ecology, uses so-called "genetic engineering"—a double misnomer. It moves genes but is not about genetics. "Engineering" implies understanding of the causal mechanisms that link actions to effects, but nobody understands the mechanisms by which genes, interacting with each other and the environment, express traits. Transgenic manipulation inserts foreign genes into random locations in a plant's DNA to see what happens. That's not engineering; it's the industrialization of life by people with a narrow understanding of it.

The results, too, are more worrisome than those of mere mechanical tinkering, because unlike mechanical contrivances, genetically modified organisms reproduce, genes spread, and mistakes literally take on a life of their own. Herbicide-resistance genes may escape to make "superweeds." Insecticide-making genes may kill beyond their intended targets. Both these problems have already occurred; their ecological effects are not yet known. Among other recent unpleasant surprises, spliced genes seem unusually likely to spread to other organisms. Canola pollen can waft spliced genes more than a mile, and common crops can hybridize with completely unrelated weeds.

Gene-spliced Bt insecticide in corn pollen kills Monarch butterflies; that insecticide, unlike its natural forebear, can build up in soil; and corn borers' resistance to it is apparently a dominant trait, so planned anti-resistance procedures won't work.

It could get worse. Division into species seems to be nature's way of keeping pathogens in a box where they behave properly (they learn that it's a bad strategy to kill your host). Transgenics may let pathogens vault the species barrier and enter new realms where they have no idea how to be-

have. It's so hard to eradicate an unwanted wild gene that we've intentionally done it only once—with the smallpox virus.

Since evolution is a fundamental process, it must occur at every scale at which it's physically possible, down to and including the nanoecosystem of the genome. It's unwise to assume, as "genetic engineers" generally do, that 90-plus percent of the genome is "garbage" or "junk" because they don't know its function. That mysterious, messy, ancient stuff is the context that influences how genes express traits. It's the genetic version of biodiversity, which in larger ecosystems is the source of resilience and endurance.

Transgenics is showing disturbing historical parallels to another problematic invention, nuclear fission. In both enterprises, technical ability has evolved faster than social institutions; skill has outrun wisdom. Both have overlooked fundamentals, often from other disciplines wrongly deemed irrelevant. Both have overreached—too far, too fast, too uncritical.

Our key choices now are not between unwelcome alternatives—nuclear power or freezing in the dark, transgenic crops or starvation—but between those bad choices and attractive ones outside the orthodoxy.

For crops, the best choice would be fairer distribution of food grown by a respectful and biologically informed agriculture that stops treating soil like dirt. But sound choices tend to emerge and get adopted in time only if we take seriously the discipline of mindful markets and the wisdom of informed democracy. Botanists have a professional duty to help us all understand the vital differences between biology and biotechnology—between the foundations of their traditional science and the scientifically immature but commercially hell-for-leather enterprise, a billion times younger, that aims to replace it.

6

A Scientific Audit: The ‘Golden Rice’— How Not to Do Science

By Dr. Mae-Wan Ho

The “golden rice”—a GE rice engineered to produce pro-vitamin A—is being offered to the Third World as cure for widespread vitamin A deficiency.

The audit uncovers fundamental deficiencies in all aspects, from the scientific/social rationale to the science and technology involved. It is being promoted in order to salvage a morally as well as financially bankrupt agricultural biotech industry.

The scientific/social rationalization for the project exposes a reductionist self-serving scientific paradigm that fails to see the world beyond its own narrow confines. The “golden rice” is a useless application. Some 70 patents have already been filed on the GE genes and constructs used in making the “golden rice.” It is a drain on public resources and a major obstruction to the implementation of sustainable agriculture that can provide the real solutions to world hunger and malnutrition.

“Golden rice” is not a “second generation” GE crop as has been claimed. It involves standard first generation technology and carries some of the worst features in terms of hazards to health and biodiversity. Rockefeller Foundation, the major funder of the project by far, has withdrawn support from it. The project should be abandoned altogether.

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A Gift-Horse for the Poor

A report in *Financial Times* states that the creators of “golden rice” have struck “a ground-breaking deal” with corporate giant AstraZeneca to give Third World farmers free access to the grain while allowing it to be commercially exploited in the developed world. The company will oversee the production of stable GE line(s) and patenting and take the lines through field trials and commercial approval. While farmers in developed countries will have to pay royalties, those in the Third World earning less than U.S. \$10,000 will not. But will Third World farmers be allowed to save the seeds for replanting? It did not say.

This “golden rice,” not yet available, is already worth its weight in diamonds. The project was funded from four sources of public finance totaling U.S. \$100 million: the philanthropic Rockefeller Foundation, whose mission is to support scientific research that specifically benefit the poor, the Swiss Federal Institute of Technology, the European Community Biotech Program, and the Swiss Federal Office for Education and Science.

The announcement failed to mention that there are already 70 patent claims on the genes, DNA sequences, and gene constructs used to make the “golden rice.” Will the cost of paying royalties for the previous 70 patent claims be added to the cost of the “golden rice”? Which of the royalties on the 70-odd patents would the Third World farmers be absolved from paying? Rockefeller Foundation, the major funder by far, has reportedly abandoned the project to “shift its agricultural funding focus to support research that will have a more direct benefit to subsistence farmers.”

The Scientific/Social Rationale Is Fallacious

Many have commented on the absurdity of offering “golden rice” as the cure for vitamin A deficiency when there are plenty of alternative, infinitely cheaper sources of vitamin A or pro-vitamin A, such as green vegetables and unpolished rice, which would be rich in other essential vitamins and minerals besides. To offer the poor and malnourished a high-tech “golden rice” tied up in multiple patents, that has cost U.S. \$100 million to produce and may cost as much to develop, is worse than telling them to eat cake.

“Golden rice” was engineered to produce pro-vitamin A or b-carotene (the substance that makes carrots orange) in the endosperm, i.e., the part of the rice grain that remains after it has been polished. The scientific paper started with a review of the literature to rationalize why such GE rice is needed and of benefit for the Third World. The paper was accompanied by an unusually long news feature entitled, “The Green Revolution Strikes Gold,” which reinforced the rationalization for the project, explaining the remarkable feat of technology involved and stated that the scientists intend to make the “golden rice” “freely available to the farmers who need it most.” The last sentence in this glowing report, however, gave the game

away: "One can only hope that this application of plant genetic engineering to ameliorate human misery without regard to short-term profit will restore this technology to political acceptability."

What were the reasons for the scientists to embark on the project? It is important to know, as these reasons may have been used to persuade funders to support the project in the first place, and funders ought to bear as much of the responsibility.

The first reason given is that the aleurone layer (in unpolished rice) is usually removed by milling as it turns rancid on storage, especially in tropical areas; and the remaining endosperm lacks pro-vitamin A. The researchers are tacitly admitting that at least some varieties of unpolished rice will have pro-vitamin A. The reason rice is milled is to prolong storage for export, and to suit the tastes of the developed world. So why not give the poor access to unpolished rice? A proportion of every rice harvest could be kept unpolished and either given freely to the poor, or sold at the cheapest prices. But the scientists have not considered that possibility. Unpolished rice is fact part of the traditional Asian diet until the Green Revolution when aggressive marketing of white polished rice created a stigma of unpolished rice. However, most rural communities still consume unpolished rice and now that consumers have become aware of its nutritional value, unpolished rice is becoming sought after.

"Predominant rice consumption," the researchers claim, promotes vitamin A deficiency, a serious health problem in at least 26 countries, including highly populated areas of Asia, Africa, and Latin America. Some 124 million children worldwide are estimated to be vitamin A deficient. (Actually, the latest figures quoted in a press release from the International Rice Research Institute (IRRI) is 250 million preschool children.) The scientists seem to be unaware that people do not eat plain rice out of choice. The poor do not get enough to eat and are undernourished as well as malnourished. The Food and Agricultural Organization (FAO) started a project in 1985 to deal with vitamin A deficiency using a combination of food fortification, food supplements, and general improvements in diets by encouraging people to grow and eat a variety of green leafy vegetables. One main discovery is that the absorption of pro-vitamin A depends on the overall nutritional status, which in turn depends on the diversity of the food consumed.

"Predominant rice consumption" is most likely to be accompanied by other dietary deficiencies. A recent study by the Global Environmental Change Programme concludes that predominant consumption of Green Revolution crops is responsible for iron deficiency in an estimated 1.5 billion, or a quarter of the world's population. The worst affected areas are in rice-growing regions in Asia and South-East Asia where the Green Revolution had been most successful in increasing crop yield.

Research institutions such as IRRI have played the key role in introduc-

ing Green Revolution crops to the Third World. IRRI was founded in 1959 under an agreement forged by the Rockefeller and Ford Foundations with the Philippine government, and its lease for operation expires in 2003. At its recent 40th anniversary celebration, hundreds of Filipino rice farmers protested against IRRI for introducing GE crops, blaming IRRI, among other things, for promoting the Green Revolution and causing massive loss of biological diversity in rice paddies throughout Asia.

It is clear that vitamin A deficiency is accompanied by deficiencies in iron, iodine, and a host of micronutrients, all of which come from the substitution of a traditionally varied diet with one based on monoculture of the Green Revolution. The real cure is to re-introduce agricultural biodiversity in the many forms of sustainable agriculture already being practiced successfully by tens of millions of farmers all over the world.

As the scientists know, clinical deficiency can be dealt with by prescription of vitamin A pills, which are affordable and immediately available. "Oral delivery of vitamin A is problematic," they state. Judging from the reference cited they may be referring to the well-known harmful effects of vitamin A overdose. But why would high levels of pro-vitamin A rice in a staple food that people generally consume in the largest amounts in a meal not also cause problems connected with overdose? In particular, vitamin A poisoning has been known to result from excessive b-carotene intake in food.

Finally, why is it necessary to genetic engineer rice? "Because no rice cultivars produce [pro-vitamin A] in the endosperm, recombinant technologies rather than conventional breeding are required." This is the conclusion to the whole fallacious reasoning process. It amounts to this: rice is polished, which removes pro-vitamin A, therefore a hundred million dollars (much of it tax-payers' money) are needed to put pro-vitamin A into polished rice. A more likely explanation is that the geneticists are looking for funding to do their research, and have constructed, as best they could, a series of rationalizations for why they should be supported. Neither the scientists nor the funders have looked further beyond the technology to people's needs and aspirations, or to what the real solutions are.

The Science and Technology Is Standard First Generation

It took ten years to engineer b-carotene into polished rice because rice naturally does not have the metabolic pathway to make it in the endosperm, perhaps for good biological reasons. Immature rice endosperm makes the early precursor, geranylgeranyl-diphosphate (GGPP). In order to turn GGPP into b-carotene, four metabolic reactions are needed, each catalyzed by a different enzyme. Enzyme 1, phytoene synthase converts GGPP to phytoene, which is colorless. Enzymes 2 and 3, phytoene desaturase and z-

carotene desaturase, each catalyzes the introduction of two double-bonds into the phytoene molecule to make lycopene, which is red in color. Finally, Enzyme 4, lycopene b-cyclase turns lycopene into b-carotene. Hereafter, the enzymes will be referred to by numbers only. Thus, a total of four enzymes have to be engineered into the rice in such a way that the enzymes are expressed in the endosperm. Some very complicated artificial gene constructs have to be made. The gene constructs are made in units called expression cassettes (see Box 1)

Box 1

The 'Gene Expression Cassette'—A Unit of Transgenic Construct
The gene for each enzyme never goes in alone. It has to be accompanied by a special piece of genetic material (DNA), the promoter, which signals the cell to turn the gene on, i.e., to transcribe the DNA gene sequence into RNA. At the end of the gene, there has to be another signal, a terminator, to mark the RNA so it can be translated into protein. To target the protein to the endosperm, an extra bit of DNA, a transit sequence, is required. The resulting expression cassette for each gene is as follows:

promoter transit sequence gene terminator

Typically, each bit of the construct: promoter, transit sequence, gene and terminator is from a different source. Several expression cassettes are usually linked in series, or “stacked” in the final construct.

In order to select for the plant cells that have taken up the foreign genes and gene-constructs, “golden rice” makes use of a standard antibiotic resistance gene coding for hygromycin resistance, also equipped with its own promoter and terminator. All these expression cassettes have to be introduced into the rice plant cells. One simplification available is that the reactions catalyzed by two of the enzymes, 2 and 3, could be done by a single bacterial enzyme, let's call it enzyme 2-3, so a total of four expression cassettes have to be introduced, one for each of three enzymes and the fourth for the antibiotic resistance marker.

Unlike natural genetic material which consists of stable combinations of genes that have co-existed for billions of years, artificial constructs consist of combinations that have never existed, not in billions of years of evolution. Artificial gene-constructs are well-known to be structurally unstable, which means they tend to break and join up incorrectly, and with other bits of genetic material, resulting in new unpredictable combinations. This process of breaking and joining of genetic material is referred to as recombination. The more complicated the construct, the more it tends to break and rearrange or form new combinations. The instability of the construct

means that it is seldom inserted into the plant genome in its intended form. The inserts are generally rearranged, with parts deleted, or repeated.

In order to make many copies of the construct and to facilitate entry into plant cells, the construct is spliced into an artificial vector, which is generally made from genetic parasites that live inside cells. The artificial vector also enables the construct to be efficiently smuggled into the plant cell and to jump into the genome of the plant cell. The vector used in the case of the “golden rice” is the one most widely used since the beginning of plant genetic engineering. It is derived from the “T-DNA,” part of the tumor-inducing (Ti) plasmid (a genetic parasite) of the soil bacterium, *Agrobacterium*. The Ti plasmid naturally invades plant cells, inserting the T-DNA into the plant cell genome, and causing the cell to develop into a plant tumor or gall. The artificial gene construct is spliced in between the left and right borders of the T-DNA vector. The borders of the T-DNA are “hotspots” for recombination, i.e., they have a pronounced tendency to break and join up, which is ultimately why the vector can invade the plant’s genome and carry its hitch-hiker gene construct along with it.

Three different constructs were made. The first consists of the expression cassettes of enzyme 1 from daffodils and enzyme 2-3 from the plant bacterial pathogen, *Erwinia uredovora*, together with the expression cassette of an antibiotic resistance marker gene that codes for hygromycin resistance. Another antibiotic resistance gene (coding for kanamycin resistance) is also present, albeit lacking a promoter. Hygromycin and kanamycin are both aminoglycoside antibiotics that inhibit protein synthesis. The resistance genes originate from bacteria and generally have specificities for more than one aminoglycoside antibiotic. This first construct is the most complicated, but it still does not have all the required enzymes. Enzyme 1 and the hygromycin resistance gene are both equipped with a promoter from the cauliflower mosaic virus (CaMV), which is especially hazardous (see below).

The second construct consists of the expression cassettes of enzyme 1 and enzyme 2-3 as in the first, but without any antibiotic resistance marker genes. The third construct consists of the expression cassette of enzyme 4, again from daffodil, stacked with the hygromycin-resistance marker-gene cassette. The strategy of separating the genes for the enzymes and antibiotic resistance marker into two different constructs is that it overcomes some of the problems of structural instability: the more cassettes stacked, the more unstable is the construct.

Each construct was spliced into a T-DNA vector, and two transformation experiments were carried out. In the first experiment, 800 immature rice embryos were inoculated with the vector containing the first construct, and hygromycin was used to select for resistant plants that have taken up the vector, resulting in 50 GE plants. In the second experiment, 500 imma-

ture embryos were inoculated with a mixture of the vectors containing the second and third construct respectively. Selection with hygromycin gave rise to 60 GE plants that have taken up the third construct, but only twelve of these had taken up the second construct as well. The transformation process is well-known to be random, as there is no way to target the foreign genes to precise locations in the genome. There could be more than one site of insertion in a single cell. Furthermore, as mentioned earlier, the actual inserts are likely to be rearranged, or subject to deletions or repetitions. Hence each transformed cell will have its own distinctive pattern of insert (s), and each GE plant, which comes from the single transformed cell, will differ from all the rest.

Note that the GE plants from the first experiment will not have the full complement of enzymes required to make b-carotene, and should give red endosperm from the lycopene present. Only the GE plants from the second experiment which have taken up both vectors would possess all the enzymes needed, and give orange-colored endosperm.

Uncontrollable Technology and Unpredictable Outcomes Raise Questions On Safety

Unexpectedly, transgenic plants from both transformation experiments gave orange polished grains. Chemical analyses confirmed that only b-carotene, in varying amounts, was found in all lines, but no lycopene. This suggests that enzyme 4 may be present in rice endosperm normally, or it could be induced by lycopene, to turn all of the lycopene into b-carotene. Lutein and zeaxanthin, two other products derived from lycopene, were also identified in varying amounts besides b-carotene. All of these were absent from non-GM rice.

In addition, many other uncharacterized, unidentified products were found, which differ from one line to another. What is the nutritional value of the other products? Are any of the known and unknown products harmful? Without thorough chemical analyses and toxicity tests, it is impossible to tell. This highlights the unpredictable, uncontrollable nature of the technology.

Molecular analyses of the GE inserts were not done in any detail. Nevertheless, judging from the evidence presented, there are the usual signs of deletions, rearrangements, and multiple repeats of the constructs inserted due to structural instability of the constructs and the tendency for recombination. There is no guarantee that any of the plants will give stable progeny in successive generations. The instability of GE lines is well-known and is a continuing problem for the industry. Inserted genes can lose their activities or become lost altogether in subsequent generations. There is nothing in “golden rice” to distinguish it from standard first generation GE plants with all the well-known defects and hazards.

‘Golden Rice’ Is No Technical Improvement And More Unsafe

“Golden rice” exhibits all the undesirable, hazardous characteristics of existing GE plants, and in added measure on account of the increased complexity of the constructs and the sources of genetic material used. The hazards are highlighted below:

- It is made with a combination of genes and genetic material from viruses and bacteria, associated with diseases in plants, and from other non-food species.
- The gene constructs are new, and have never existed in billions of years of evolution.
- Unpredictable by-products have been generated due to random gene insertion and functional interaction with host genes, which will differ from one plant to another.
- Over-expression of transgenes linked to viral promoters, such as that from CaMV, exacerbates unintended metabolic effects as well as instability (see below). There are at least two CaMV promoters in each transgenic plant of the “golden rice,” one of which is linked to the antibiotic resistance marker gene.
- The transgenic DNA is structurally unstable, leading to instability of the GE plants in subsequent generations, multiplying unintended, random effects.
- Structural instability of transgenic DNA increases the likelihood of horizontal gene transfer and recombination.
- Instability of transgenic DNA is enhanced by the CaMV promoter, which has a recombination hotspot, thereby further increasing the potential for horizontal gene transfer.
- The CaMV promoter is promiscuous in function and works efficiently in all plants, in green algae, yeast and *E. coli*. The spread of genes linked to this promoter by ordinary cross-pollination or by horizontal gene transfer will have enormous impacts on health and biodiversity. In particular, the hygromycin resistance gene linked to it may be able to function in bacteria associated with infectious diseases.
- Horizontal transfer of transgenic DNA from GE plants into soil fun-

gi and bacteria has been demonstrated in laboratory experiments. Recent evidence suggests that it has also taken place in a field-trial site for GE sugar-beets, in which transgenic DNA persisted in the soil for at least two years afterwards.

- Prof. Hans-Hinrich Kaatz from the University of Jena, has just presented new evidence of horizontal gene transfer within the gut of bee larvae. Pollen from GE rapeseed tolerant to the herbicide glufosinate were fed to immature bee larvae. When the microorganisms were isolated from the gut of the larvae and examined for the presence of the gene conferring glufosinate resistance, it was found in some of the bacteria as well yeast cells.
- All cells including those of human beings are now known to take up genetic material. While natural (unmanipulated) genetic material is simply broken down to supply energy, invasive pieces of genetic material may jump into the genome to mutate genes. Some insertions of foreign genetic material may also be associated with cancer.
- Horizontal transfer of genes and constructs from the “golden rice” will spread transgenes, including antibiotic resistance genes, to bacterial pathogens, and also has the potential to create new viruses and bacteria associated with diseases.

Conclusion

In conclusion, the “golden rice” project was a useless application, a drain on public finance and a threat to health and biodiversity. It is being promoted in order to salvage a morally as well as financially bankrupt agricultural biotech industry, and is obstructing the essential shift to sustainable agriculture that can truly improve the health and nutrition especially of the poor in the Third World. This project should be terminated immediately before further damage is done.

The “golden rice” possesses all the usual defects of first generation transgenic plants plus multiple copies of the CaMV promoter which we have strongly recommended withdrawing from use on the basis of scientific evidence indicating this promoter to be especially unsafe. A growing number of scientists (318 scientists from 39 countries to date) are calling for a global moratorium on the environmental releases of GEOs until and unless they can be shown to be safe.

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7

Report from Seattle: The WTO Meets Skeleton Woman

By Paul Hawken

When I was able to open my eyes, I saw lying next to me a young man, 19, maybe 20 at the oldest. He was in shock, twitching and shivering uncontrollably from being tear-gassed and pepper-sprayed at close range. His burned eyes were tightly closed, and he was panting irregularly. Then he passed out. He went from excruciating pain to unconsciousness on a sidewalk wet from the water that a medic had poured over him to flush his eyes--like a young boy in bed.

This is what I remember about the violence. There was almost none until police attacked demonstrators that Tuesday in Seattle. Michael Meacher, Environment Minister of the United Kingdom, said afterward, "What we hadn't reckoned with was the Seattle Police Department who single-handedly managed to turn a peaceful protest into a riot." There was no police restraint, despite what Mayor Paul Schell kept proudly assuring television viewers all day. Instead, there were rubber bullets, which Schell kept denying all day. In the end, more copy and video was given to broken windows than broken teeth.

As I tried to find my way down Sixth Street after the tear gas and pepper spray, I couldn't see. Anita Roddick found and guided me. When your eyes fail, your ears take over. I could hear acutely. What I heard was anger, dismay, shock. For many people, including the police, this was their first direct action. Demonstrators who had taken non-violent training were astonished at the police brutality. I heard young voices, incredulous, stunned.

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The demonstrators were students, their professors, clergy, lawyers, and medical personnel. They held signs against Burma and violence. They dressed as butterflies.

More than 1,500 non-governmental organizations registered with the World Trade Organization. More than 700 organizations and between 40,000 and 60,000 people took part in the protests. These groups and citizens sense a cascading loss of human and labor rights in the world. Seattle was not the beginning but simply the most striking expression of citizens struggling against a worldwide corporate-financed oligarchy—in effect, a plutocracy. Oligarchy and plutocracy are not polite terms. They often are used to describe “other” countries where a small group of wealthy people rule, but not the “first world”—the United States, Japan, Germany, or Canada. The World Trade Organization, however, is trying to cement into place that corporate plutocracy. Already, the world’s top 200 companies have twice the assets of 80 percent of the world’s people. And this polarization and concentration of wealth is increasing. Global corporations represent a new empire whether they are aware of it or not. With massive amounts of capital at their disposal, any of which can be used to influence politicians and the public as and when deemed necessary, they threaten and diminish all democratic institutions are diminished and at risk. Corporate free market policies subvert culture and community, a true tyranny. The American Revolution occurred because of crown-chartered corporate abuse, a “remote tyranny” in Thomas Jefferson’s words. To see Seattle as a singular event, as did most of the media, is to look at the battles of Concord and Lexington as meaningless skirmishes.

But the mainstream media, consistently problematic in their coverage of any type of protest, had an even more difficult time understanding and covering both the issues and activists in Seattle. No charismatic leader led. No religious figure engaged in direct action. No movie stars starred. There was no alpha group. The Ruckus Society, Rainforest Action Network, Global Exchange, and hundreds more were there, coordinated primarily by cell phones, emails, and the Direct Action Network. They were up against the Seattle Police Department, the Secret Service, and the FBI—to say nothing of the media coverage and the WTO itself.

Thomas Friedman, *New York Times* columnist and author of an elegy to globalization entitled *The Lexus and the Olive Tree*, angrily wrote that the demonstrators were “a Noah’s ark of flat-earth advocates, protectionist trade unions and yuppies looking for their 1960s fix.” Not so. They were organized, educated, and determined. They were human rights activists, labor activists, indigenous people, people of faith, steel workers, and farmers. They were forest activists, environmentalists, social justice workers, students, and teachers. And they wanted the World Trade Organization to listen. They were speaking on behalf of a world that has not been made better

by globalization. Income disparity is growing rapidly. The difference between the top and bottom quintiles has doubled in the past 30 years. Eighty-six percent of the world's goods go to the top 20 percent, the bottom fifth get 1 percent. The apologists for globalization cannot support their contention that open borders, reduced tariffs, and forced trade benefit the poorest three billion people in the world. Globalization does, however, create the concentrations of capital seen in northern financial and industrial centers--indeed, the wealth in Seattle itself. Since the people promoting globalized free trade policies live in those cities, it is natural that they should be biased. Despite Friedman's invective about "the circus in Seattle," the demonstrators and activists who showed up there are not against trade. They do demand proof that shows when and how trade--as the WTO constructs it--benefits workers and producers abroad, as well as workers in developing nations. And that proof is simply non-existent.

Earlier that day, November 30, I had walked toward the Convention Center with Randy Hayes, the founder of Rainforest Action Network. As soon as we turned the corner on First Street and Pike Avenue, we could hear drums, chants, sirens, roars. As we approached Fifth, police stopped us. We could go no farther without credentials. Ahead of us were thousands of protesters. Beyond them was a large cordon of gas-masked and riot-shielded police, an armored personnel carrier, and fire trucks. On one corner was Niketown. On the other, the Sheraton Hotel, through which there was a passage to the Convention Center. The cordon of police in front of us tried to prevent more protesters from joining those who blocked the entrances to the Convention Center. Randy was a credentialed WTO delegate. He showed his pass to the officer who thought it looked like me. The officer joked with us, kidded Randy about having my credential and then winked and let us both through. The police were still relaxed at that point. Ahead of us crowds were milling and moving. Anarchists were there, maybe 40 in all, dressed in black pants, black bandanas, black balaclavas, and jackboots, one of two groups identifiable by costume. The other was a group of 300 children who had dressed brightly as turtles in the Sierra Club march the day before. The costumes were part of a serious complaint against the WTO. When the United States attempted to block imports of shrimp caught in the same nets that capture and drown 150,000 sea turtles each year, the WTO called the block "arbitrary and unjustified." Thus far in every environmental dispute that has come before the WTO, its three-judge panels, which deliberate in secret, have ruled for business, against the environment. The panel members are selected from lawyers and officials who are not educated in biology, the environment, social issues, or anthropology.

Opening ceremonies for the World Trade Organization's Third Ministe-

rial were to have been held that Tuesday morning at the Paramount Theater near the Convention Center. Police had ringed the theater with Metro buses touching bumper to bumper. The protesters surrounded the outside of that steel circle. Only a few hundred of the 5,000 delegates made it inside, as police were unable to provide safe corridors for members and ambassadors. The theater was virtually empty when U.S. trade representative and meeting co-chair Charlene Barshevsky was to have delivered the opening keynote. Instead, she was captive in her hotel room a block from the meeting site. WTO Executive Director Michael Moore was said to have been apoplectic.

Mayor Paul Schell stood despondently near the stage. Since no scheduled speakers were present, Kevin Danaher, Medea Benjamin, and Juliet Hill from Global Exchange went to the lectern and offered to begin a dialogue in the meantime. The WTO had not been able to come to a pre-meeting consensus on the draft agenda. The NGO community, however, had drafted a consensus agreement about globalization--and the three thought this would be a good time to present it, even if the hall had only a desultory number of delegates. Although the three were credentialed WTO delegates, the sound system was quickly turned off and the police arm-locked and handcuffed them. Medea's wrist was sprained. All were dragged off stage and arrested. It mirrored how the WTO has operated since its birth in 1995. Listening to people is not its strong point. WTO rules runs roughshod over local laws and regulations. It relentlessly pursues the elimination of any strictures on the free flow of trade, including how a product is made, by whom it is made, or what happens when it is made. By doing so, the WTO is eliminating the ability of countries and regions to set standards, to express values, or to determine what they do or don't support. Child labor, prison labor, forced labor, substandard wages and working conditions cannot be used as a basis to discriminate against goods. Nor can environmental destruction, habitat loss, toxic waste production, and the presence of transgenic materials or synthetic hormones cannot be used as the basis to screen or stop goods from entering a country. Under WTO rules, the Sullivan Principles and the boycott of South Africa would not have existed. If the world could vote on the WTO, would it pass? Not one country of the 135-member states of the WTO has held a plebiscite to see if their people support this concept. The people trying to meet in the Green Rooms at the Seattle Convention Center were not elected. Even Michael Moore was not elected.

But while the Global Exchange was temporarily silenced, the Direct Action Network's plan was working brilliantly on the outside of the Convention Center. The plan was simple: insert groups of trained non-violent activists into key points downtown, making it impossible for delegates to move. DAN had hoped that 1,500 people would show up. Close to 10,000

did. The 2,000 people who began the march to the Convention Center at 7 a.m. from Victor Steinbrueck Park and Seattle Central Community College were composed of affinity groups and clusters whose responsibility was to block key intersections and entrances. Participants had trained for many weeks in some cases, for many hours in others. Each affinity group had its own mission and was self-organized. The streets around the Convention Center were divided into 13 sections and individual groups and clusters were responsible holding these sections. There were also “flying groups” that moved at will from section to section, backing up groups under attack as needed. The groups were further divided into those willing to be arrested, and those who were not. As protesters were beaten, gassed, clubbed, and pushed back, a new group would replace them. Throughout most of the day, using a variety of techniques, groups held intersections and key areas downtown. The protests were organized through a network of cell phones, bullhorns, and signals. All decisions prior to the demonstrations were reached by consensus. Minority views were heeded and included. The one agreement shared by all was no violence, physical or verbal, no weapons, no drugs or alcohol. There were no charismatic leaders barking orders. There was no command chain. There was no one in charge. Police said that they were not prepared for the level of violence, but in fact they were unprepared for a network of non-violent protesters totally committed to one task—shutting down the WTO.

Moore and Barshevsky’s frustration was shared by Madeleine Albright, the Clinton advance team, and chief of staff John Podesta. This was to have been a celebration, a victory, one of the crowning achievements to showcase the Clinton administration, the moment when it would consolidate its centrist free trade policies, allowing the Democrats to show multinational corporations that they could deliver the goods. This was to have been Barshevsky’s moment, an event that would give her the inside track to become Secretary of Commerce in the Gore Administration. This was to have been Michael Moore’s moment, reviving what had been a mediocre political ascendancy in New Zealand. To say nothing of Monsanto’s moment. If the as-yet unapproved draft agenda were ever ratified, the Europeans could no longer block or demand labeling on genetically modified crops without being slapped with punitive lawsuits and tariffs. The draft also contains provisions that would allow all water in the world to be privatized. It would allow corporations patent protection on all forms of life, even genetic material in cultural use for thousands of years. Farmers who have spent thousands of years growing crops in a valley in India could, within a decade, be required to pay for their water. They could also find that they would have to purchase seeds containing genetic traits their ancestors developed, from companies that have engineered the seeds not to reproduce unless the farmer annually buys expensive chemicals to restore seed viability.

ty. If this happens, the CEOs of Novartis and Enron, two of the companies creating the seeds and privatizing the water, will have more money. What will Indian farmers have?

But the perfect moment for Barshevsky, Moore, and Monsanto didn't arrive. The meeting couldn't start. Demonstrators were everywhere. Private security guards locked down the hotels. The downtown stores were shut. Hundreds of delegates were on the street trying to get into the Convention Center. No one could help them. For WTO delegates accustomed to an ordered corporate or governmental world—it was a calamity.

Up Pike toward Seventh and to Randy's and my right on Sixth, protesters faced armored cars, horses, and police in full riot gear. In between, demonstrators ringed the Sheraton to prevent an alternative entry to the Convention Center. At one point, police guarding the steps to the lobby pummeled and broke through a crowd of protesters to let eight delegates in. On Sixth Street, Sergeant Richard Goldstein asked demonstrators seated on the street in front of the police line "to cooperate" and move back 40 feet. No one understood why, but that hardly mattered. No one was going to move. He announced that "chemical irritants" would be used if they did not leave. The police were anonymous, black ghosts. No facial expressions, no face. You could not see their eyes. They were masked Hollywood caricatures burdened with 60 to 70 pounds of weaponry. These were not the men and women of the 6th precinct. They were the Gang Squads and the SWAT teams of the Tactical Operations Divisions, closer in training to soldiers from the School of the Americas than local cops on the beat. Behind them and around were special forces from the FBI, the Secret Service, even the CIA.

The police were almost motionless. They were equipped with U.S. military standard M40A1 double canister gas masks; uncalibrated, semi-automatic, high velocity Autocockers loaded with solid plastic shot; Monadnock disposable plastic cuffs, Nomex slash-resistant gloves, Commando boots, Centurion tactical leg guards, combat harnesses, DK5-H pivot-and-lock riot face shields, black Monadnock P24 polycarbonate riot batons with TrumBull stop side handles, No.2 continuous discharge CS (ortho-chlorobenzylidene-malononitrile) chemical grenades, M651 CN (chloroacetophenone) pyrotechnic grenades, T16 Flameless OC Expulsion Grenades, DTCA rubber bullet grenades (Stingers), M-203 (40mm) grenade launchers, First Defense MK-46 Oleoresin Capsicum (OC) aerosol tanks with hose and wands, .60 caliber rubber ball impact munitions, lightweight tactical Kevlar composite ballistic helmets, combat butt packs, 30 cal. thirty-round mag pouches, and Kevlar body armor. None of the police had visible badges or forms of identification.

The demonstrators seated in front of the black-clad ranks were equipped with hooded jackets for protection against rain and chemicals. They carried

toothpaste and baking powder for protection of their skin, and wet cotton cloths impregnated with vinegar to cover their mouths and noses after a tear-gas release. In their backpacks were bottled water and food for the day ahead.

Ten Koreans came around the corner carrying a 10-foot banner protesting genetically modified foods. They were impeccable in white robes, sashes, and headbands. One was a priest. They played flutes and drums and marched straight toward the police and behind the seated demonstrators. Everyone cheered at the sight and chanted "The whole world is watching." The sun broke through the gauzy clouds. It was a beautiful day. Over cell phones, we could hear the cheers coming from the labor rally at the football stadium. The air was still and quiet. We waited.

At 10 a.m. the police fired the first seven canisters of tear gas into the crowd. The whitish clouds wafted slowly down the street. The seated protesters were overwhelmed, yet most did not budge. Police poured over them. Then came the truncheons, and the rubber bullets. I was standing with a couple hundred people who had ringed the hotel, arms locked. We watched as long as we could until the tear gas slowly enveloped us. We were several hundred feet from Sgt. Goldstein's 40-foot "cooperation" zone. Police pushed and truncheoned their way through and behind us. We had covered our faces with rags and cloth, snatching glimpses of the people being clubbed in the street before shutting our eyes. The gas was a fog through which people moved in slow, strange dances of shock and pain and resistance. Tear gas is a misnomer. Think about feeling asphyxiated and blinded. Breathing becomes labored. Vision is blurred. The mind is disoriented. The nose and throat burn. It's not a gas, it's a drug. Gas-masked police hit, pushed, and speared with the butt ends of their batons. We then sat down, hunched over, and locked arms more tightly. By then, the tear gas was so strong our eyes couldn't open. One by one, our heads were jerked back from the rear, and pepper was sprayed directly into each eye. It was very professional. Like hair spray from a stylist. Sssst. Sssst.

Pepper spray is derived from cayenne peppers. It is food-grade, pure enough to be used in salsa. The spray used in Seattle is the strongest available, containing 10 percent to 15 percent Oleoresin Capsicum, with a 1.5 to 2.0 million Scoville heat unit rating. One to three Scoville units are when your tongue can first detect hotness. (The jalapeno pepper is rated between 2,500 to 5,000 Scoville units. The habanero, usually considered the hottest pepper in the world, is rated around 300,000 Scoville units.) This description was written by a police officer who sells pepper spray on his website. It is about his first experience being sprayed during a training exercise:

"The pepper spray stream then hit my eyes. It then felt as if two red-hot pieces of steel were grinding into my eyes, as if someone was blowing a red-hot cutting torch into my face. I then fell to the ground just like all the

others and started to rub my eyes even though I knew better not to. The heat from the pepper spray was overwhelming. I could not resist trying to rub it off of my face. The pepper spray caused my eyes to shut very quickly. The only way I could open them was by prying them open with my fingers. Everything that we had been taught about pepper spray had turned out to be true. And everything that our instructor had told us that we would do, even though we knew not to do it, we still did. Pepper spray turned out to be more than I had bargained for.”

The Seattle Police had made a decision not to arrest people. Throughout the day, the affinity groups created through Direct Action stayed together. Tear gas, rubber bullets, and pepper spray were used so much that by late afternoon, supplies ran low. What seemed like an afternoon lull or standoff was because police had used up all their stores. Officers combed surrounding counties for tear gas, sprays, concussion grenades, and munitions. As police restocked, the word came down from the White House to secure downtown Seattle or the WTO meeting would be called off. By late afternoon, the Mayor and Chief announced a 7 p.m. curfew, “no protest” zones, and declared the city under civil emergency. The police were fatigued and frustrated. Over the next seven hours and into the night, the police turned downtown Seattle into Beirut.

That morning, it was the police commanders that were out of control, ordering the gassing and pepper spraying and shooting of people protesting non-violently. By evening, it was the individual police who were out of control. Anger erupted, protesters were kneed and kicked in the groin, and police used their thumbs to grind the eyes of pepper-spray victims. A few demonstrators danced on burning dumpsters that were ignited by pyrotechnic tear-gas grenades (the same ones used in Waco). Taunting, jeering, protesters were defiant. Tear gas canisters were being thrown back as fast as they were launched. Drum corps marched using empty 5-gallon water bottles for instruments. Despite their steadily dwindling number, maybe 1,500 by evening, a hardy number of protesters held their ground, seated in front of heavily armed police, hands raised in peace signs, submitting to tear gas, pepper spray, and riot batons. As they retreated to the medics, new groups replaced them. Every channel covered the police riots live. On TV, the police looked absurd, frantic, and mean. No one could believe what they were seeing. Passing Metro buses filled with passengers were gassed. Police were pepper spraying residents and bystanders. The Mayor went on TV that night to say, that as a protester from the '60s, he never could have imagined what he was going to do next: Call in the National Guard.

During that day, the anarchist black blocs were in full view. Numbering about one hundred, they could have been arrested at any time but the police were so weighed down by their own equipment, they literally couldn't run. Both the police and the Direct Action Network had mutually apprised each

other for months prior to the WTO about the anarchists' intentions. The Eugene Police had volunteered information and specific techniques to handle the black blocs, but had been rebuffed by the Seattle Police. It was widely known they would be there, and that they had property damage in mind. To the credit of the Mayor, the Police Chief, and the Seattle press, distinctions were consistently made between the protesters and the anarchists (later joined by local vandals as the night wore on). But the anarchists were not primitivists, nor were they from Eugene. They were well organized, and they also had a plan.

The black blocs came with tools (crowbars, hammers, acid-filled eggs) and hit lists. They knew they were going after Fidelity Investments but not Charles Schwab. Starbucks but not Tully's. The GAP but not REI. Fidelity Investments because they are large investors in Occidental Petroleum, the oil company most responsible for the violence against the U'wa tribe in Columbia. Starbucks because of their non-support of fair-traded coffee. The GAP because of the Fisher family's purchase of Northern California forests. They targeted multinational corporations whom they see as benefiting from repression, exploitation of workers, and low wages. According to one anarchist group, the ACME collective: "Most of us have been studying the effects of the global economy, genetic engineering, resource extraction, transportation, labor practices, elimination of indigenous autonomy, animal rights and human rights and we've been doing activism on these issues for many years. We are neither ill-informed nor inexperienced." They don't believe we live in a democracy, do believe that property damage (windows and tagging primarily) is a legitimate form of protest, and that it is not violent unless it harms or causes pain to a person. For the black blocs, breaking windows is intended to break the spells cast by corporate hegemony, an attempt to shatter the smooth exterior facade that covers corporate crime and violence. That's what they did. And what the media did is what I just did in the last two paragraphs: Report on the desires and recount the property damage caused by a tiny sliver of the 40,000 marchers and demonstrators.

It's not inapt to compare the carefully considered lawlessness of the anarchists with the equally carefully considered flouting of other laws by the WTO. When the "The Final Act Embodying the Results of the Uruguay Round of Multilateral Trade Negotiations" was enacted April 15th, 1994 in Marrakech, it was recorded as a 550-page agreement that was then sent to Congress for passage. Ralph Nader offered to donate \$10,000 to any charity of a congressman's choice if any of them signed an affidavit saying they had read it and could answer several questions about it. Only one congressman--Sen. Hank Brown, a Colorado Republican--took him up on it. After reading the document, Brown changed his opinion and voted against the Agreement. There were no public hearings, dialogue, or education. What passed

is an Agreement that gives the WTO the ability to overrule prior U.S. conventions, acts, treaties, and agreements. The WTO directly violates “The Universal Declaration of Human Rights” adopted by member nations of the United Nations, not to mention Agenda 21, the Convention on Biodiversity, and others. Most of the delegates to Marrakech, even the heads of country delegations, were not included or made aware of the WTO statutes that were drafted by sub-groups of bureaucrats and lawyers, some of whom represented transnational corporations.

The police mandate to clear downtown was achieved by 9 p.m. Tuesday night. But police, some who were fresh recruits from outlying towns, didn’t want to stop there. They chased demonstrators into neighborhoods where the distinctions between protesters and citizens vanished. The police began attacking bystanders, witnesses, residents, and commuters. They had completely lost control. When President Clinton sped from Boeing airfield to the Westin at 1:30 a.m. Wednesday, his limousines entered a police-ringed city of broken glass, helicopters, and boarded windows. He was too late. The mandate for the WTO had vanished sometime that afternoon.

Over the next few days, a surprised press corps went to work and spun webs. They created myths, told fables. They vented thinly veiled anger in columns, and pointed guilt-mongering fingers at brash, spoiled white kids who did not understand the issues. Supposedly, anarchists led by anarcho-primitive John Zerzan from Eugene ran rampant. Misguided demonstrators held self-canceling views. Protesters were against trade. Patricia King, one of two *Newsweek* reporters in Seattle, called me from her hotel room at the Four Seasons and wanted to know if this was the ’60s redux. No, I told her. The ’60s were primarily an American event; the protests against the WTO are international. Who are the leaders? she wanted to know. There are no leaders in the traditional sense. But there are thought leaders, I said. Who are they? she wanted to know. I began to name some, including their writings, area of focus, and organizational affiliations: Martin Khor and Vandana Shiva of the Third World Network, Walden Bello of Focus on the Global South, Maude Barlow of the Council of Canadians, Tony Clarke of Polaris Institute, Jerry Mander of the IFG, Susan George of the Transnational Institute, David Korten of the People-Centered Development Forum, John Cavanagh of the Institute for Policy Studies, Lori Wallach of Public Citizen of the International Society for Ecology and Culture, Mark Ritchie of the Institute For Agriculture and Trade Policy, Anuradha Mittal of Institute for Food & Development Policy, Helena Norberg-Hodge of the International Society for Ecology and Culture, Owens Wiwa of the Movement for the Survival of the Ogoni People, Chakravarthi Raghavan of the Third World Network in Geneva, Debra Harry of the Indigenous Peoples Coalition Against Biopiracy, Jose Bove of the Confederation Paysanne Eu-

rop'enne, Tetteh Hormoku of the Third World Network in Africa, Randy Hayes of Rainforest Action Network. Stop, stop, she said. I can't use these names in my article. Why not? Because Americans have never heard of them. Instead, *Newsweek* editors put the picture of the Unabomber, Theodore Kaczynski, in the article because he had, at one time, purchased some of John Zerzan's writings.

Between 40,000 and 60,000 people came to Seattle to demonstrate. What a majority of media projected onto the marchers and activists, in an often-contradictory manner, was that the protesters are afraid of a world without walls; that they want the WTO to have even more rules; that they blame the WTO for the world's problems; that they are opposed to global integration; that they have been duped by Pat Buchanan; that they are against trade; that they are ignorant and insensitive to the world's poor; that they want to tell other people how to live—the list is long and tendentious.

Some of the mainstream media also assigned blame to the protesters for the meeting's outcome. But ultimately, it was not on the streets that the WTO broke down. It was inside. It was a heated and rancorous Ministerial, and the meeting ended in a stalemate, with African, Caribbean, and some Asian countries refusing to support a draft agenda that had been negotiated behind closed doors without their participation. With that much contention inside and out, one can rightly ask whether the correct question is being posed. The question, as propounded by corporations, is how to make trade rules more uniform. The proper question, it seems to me, is how do we make trade rules more differentiated so that different cultures, cities, peoples, places, and countries benefit the most.

"Civilizations in decline are consistently characterized by a tendency toward standardization and uniformity. Conversely, during the growth stage of civilization, the tendency is toward differentiation and diversity."

— Arnold Toynbee, *A Study of History*

Those who marched and protested opposed globalization but they did not necessarily oppose internationalization of trade. Economist Herman Daly has long made the distinction between the two. Internationalization means trade between nations. Globalization refers to a system where there are uniform rules for the entire world, a world in which capital and goods move at will without the rule of individual nations. Nations, for all their faults, set trade standards. Those who are willing to meet those standards can do business with them. Do nations abuse this? Always and constantly, the U.S. being the worst offender. But it does provide, where democracies prevail, a means for people to set their own policy, to influence decisions, and determine their future. Globalization supersedes the nation, the state, the region, and the village. While eliminating nations may indeed be a

good idea, the elimination of sovereignty is not.

One recent example is that of Chiquita Brands, which recently made a large donation to the Clinton administration after the United States filed a complaint with the WTO against the European Union because European import policies favored bananas coming from small Caribbean growers. There was no question about the policies: they restricted imports from large multinational companies in Central America (plantations whose lands were secured by U.S. military force during the past century), and favored small family farmers who used fewer chemicals. It seemed like a decent thing to do, and everyone thought the bananas tasted better. For the banana giants, this was untenable. The United States prevailed in this WTO-arbitrated case. So who really won, and who lost? The self-sufficient farmers who were making a decent living prior to the decision didn't win. Did the Central American employees at Chiquita Brands win? Ask the hundreds of workers in Honduras who were made infertile by the use of Dibromochloropropane on the banana plantations. Ask the mothers whose children have birth defects from pesticide poisoning.

Globalization leads to the concentration of wealth inside large multinational corporations such as Time-Warner, Microsoft, GE, Exxon, and Wal-Mart. These giants can obliterate social capital and local equity, and create cultural homogeneity in their wake. Countries as different as Mongolia, Bhutan, and Uganda will have no choice but to allow Blockbuster, Burger King, and Pizza Hut to operate within their borders. Even Martha's Vineyard's refusal to allow a McDonald's could be nullified under the WTO regulations. The as-yet unapproved draft agenda calls for all governments to open up their procurement process to multi-national foreign corporations. No longer could local governments buy preferentially from local vendors. It could force governments to essentially privatize health and allow foreign companies to bid on delivering national health care programs. It could privatize and commodify education, or ban cultural restrictions to entertainment, advertising, or commercialism as a trade barrier. In addition, globalization kills self-reliance, since smaller local businesses can rarely compete with highly capitalized firms who seek market share instead of profits. Thus, developing regions may become more subservient to distant companies, with more of their income exported rather than being re-spent locally.

On the weekend prior to the WTO meeting, the International Forum on Globalization held a two-day teach-in at Benaroya Hall in downtown Seattle on just such questions of how countries can maintain autonomy in the face of globalization. Chaired by IFG President Jerry Mander, more than 2,500 people from around the world attended. A similar number were turned away. It was the hottest ticket in town (but somehow that ticket did not get into the hands of pundits and columnists. It was an extravagant dis-

play of research, intelligence, and concern, expressed by scholars, diplomats, writers, academics, fishermen, scientists, farmers, geneticists, businesspeople, and lawyers. Beyond and before the teach-in, non-governmental organizations, institutes, public interest law firms, farmers organizations, unions, and councils had been issuing papers, communiques, press releases, books, and pamphlets for years. They were almost entirely ignored by the WTO.

But something else was happening in Seattle underneath the debates and protests. In Stewart Brand's new book, *The Clock of the Long Now—Time and Responsibility*, he discusses what makes a civilization resilient and adaptive. Scientists have studied the same question about ecosystems. How does a system, be it cultural or natural, manage change, absorb shocks, and survive especially when change is rapid and accelerating? The answer has much to do with time, both our use of it and our respect for it. Biological diversity in ecosystems buffers against sudden shifts because different organisms and elements fluctuate at different time scales (flowers, fungi, spiders, trees, laterite, and foxes) all have different rates of change and response. Some respond quickly, other slowly, so that the system, when subjected to stress, can move, sway, and give, and then return and restore.

The WTO was a clash of chronologies or time frames, at least three, probably more. The dominant time frame was commercial. Businesses are quick, welcome innovation in general, and have a bias for change. They need to grow more quickly than ever before. They are punished, pummeled and bankrupted if they do not. With worldwide capital mobility, companies and investments are rewarded or penalized instantly by a network of technocrats and money managers who move \$2 trillion dollars a day seeking the highest return on capital. The Internet, greed, global communications, and high-speed transportation are all making businesses move faster than before.

The second time frame is culture. It moves more slowly. Cultural revolutions are resisted by deeper, historical beliefs. The first institution to blossom under *perestroika* was the Russian Orthodox Church. I walked into a church near Boris Pasternak's dacha in 1989 and heard priests and babushkas reciting the litany with perfect recall as if seventy-two years of repression had never happened. Culture provides the slow template of change within which family, community, and religion prosper. Culture provides identity and in a fast-changing world of displacement and rootlessness, becomes ever more important. In between culture and business is governance, faster than culture, slower than commerce.

At the heart, the third and slowest chronology is earth, nature, the web of life. As ephemeral as it may seem, it is the slowest clock ticking, always there, responding to long, ancient evolutionary cycles that are beyond civilization.

These three chronologies conflict. As Stewart Brand points out, business unchecked becomes crime. Look at Russia. Look at Microsoft. Look at history. What makes life worthy and allows civilizations to endure are all the things that have “bad” payback under commercial rules: infrastructure, universities, temples, poetry, choirs, literature, language, museums, terraced fields, long marriages, line dancing, and art. Most everything we hold valuable is slow to develop, slow to learn, and slow to change. Commerce requires the governance of politics, art, culture, and nature, to slow it down, to make it heedful, to make it pay attention to people and place. It has never done this on its own. The extirpation of languages, cultures, forests, and fisheries is occurring worldwide in the name of speeding up business. Business itself is stressed out of its mind by rapid change. The rate of change is unnerving to all, even to those who are benefiting. To those who are not benefiting, it is devastating. What marched in the streets of Seattle? Slower time strode into the WTO. Ancient identity emerged. The cloaks of the forgotten paraded on the backs of our children. It is not the fast things that will prevail. In the end, that which is slow is powerful.

What appeared in Seattle were the details, dramas, stories, peoples, and puppet creatures that had been ignored by the bankers, diplomats, and the rich. Corporate leaders are certain they have discovered a treasure of immeasurable value, a trove so great that surely we will all benefit. It is the treasure of unimpeded commerce flowing everywhere as fast as is possible. It is like romantic love. Bright, shining, perfect, and unassailable. But in Seattle, quick time met slow time. The turtles, farmers, and priests weren't invited and don't need to be because they are the shadow world that cannot be overlooked, that will tail and haunt the WTO, and all its successors, for as long as it exists. They will be there even if they meet in totalitarian countries where free speech is criminalized. They will be there in dreams of delegates high in the Four Seasons Hotel. They will haunt the public relations flacks who solemnly insist that putting the genes of scorpions into our food is a good thing. What gathered around the Convention Center and hotels was everything the WTO left behind.

In the Inuit tradition, there is a story of a fisherman who trolls an inlet. When a heavy pull on the fisherman's line drags his kayak to sea, he thinks he has caught the “big one,” a fish so large he can eat for weeks, a fish so fat that he will prosper ever after, a fish so amazing that the whole village will wonder at his prowess. As he imagines his fame and coming ease, what he reels up is Skeleton Woman, a woman flung from a cliff and buried long ago, a fish-eaten carcass resting at the bottom of the sea that is now entangled in his line. Skeleton Woman is so tangled in his fishing line that she is dragged behind the fisherman wherever he goes. She is pulled across the water, over the beach, and into his house where he collapses in terror. In

the retelling of this story by Clarissa Pinkola Estes, the fisherman has brought up a woman who represents life and death, a specter who reminds us that with every beginning there is an ending, for all that is taken, something must be given in return, that the earth is cyclical and requires respect. The fisherman, feeling pity for her, slowly disentangles her, straightens her bony carcass, and finally falls asleep. During the night, Skeleton Woman scratches and crawls her way across the floor, drinks the tears of the dreaming fisherman, and grows anew her flesh and heart and body. This myth applies to business as much as it does to a fisherman. The apologists for the WTO want more-engineered food, sleeker planes, computers everywhere, golf courses that are preternaturally green. They see no limits; they know of no downside. But Life always comes with Death, with a tab, a reckoning. They are each other's consorts, inseparable and fast. These expansive dreams of the world's future wealth were met with perfect symmetry by Bill Gates, Jr. the co-chair of the Seattle host committee, the world's richest man. But Skeleton woman also showed up in Seattle, the uninvited guest, and the illusion of wealth, the imaginings of unfettered growth and expansion, became small and barren in the eyes of the world. Dancing, drumming, ululating, marching in black with a symbolic coffin for the world, she wove through the sulphurous rainy streets of the night. She couldn't be killed or destroyed, no matter how much gas or pepper spray or rubber bullets were used. She kept coming back and sitting in front of the police and raised her hands in the peace sign, and was kicked, and trod upon, and it didn't make any difference. Skeleton Woman told corporate delegates and rich nations that they could not have the world. It is not for sale. The illusions of world domination have to die, as do all illusions. Skeleton Woman was there to say that if business is going to trade with the world, it has to recognize and honor the world, her life and her people. Skeleton Woman was telling the WTO that it has to grow up and be brave enough to listen, strong enough to yield, courageous enough to give. Skeleton Woman has been brought up from the depths. She has regained her eyes, voice and spirit. She is about in the world and her dreams are different. She imagines a world where children do not live on streets; she believes that the right to self-sufficiency is a human right; she imagines a world where the means to kill people is not a business but a crime, where families do not starve, where fathers can work, where children are never sold, where women cannot be impoverished because they are mothers and not whores. She cannot see in any dream a time where a man holds a patent to a living seed, or animals are factories, or people are enslaved to money, or water belongs to a stockholder. Hers are deep dreams from slow time. She is patient. She will not be quiet or flung to sea anytime soon.

8

Lundberg Family Farms' Position on Genetically Modified Organisms (GMOs)

Lundberg Family Farms opposes the use of GMOs on farms and in food because:

- They have not been proven safe for people and the environment.
- They have the ability to spread and contaminate neighboring fields with GMO seed and cause GMO crosses through the movement of pollen.
- They promote the dependence of farmers on chemicals and chemical suppliers.

Lundberg Family Farms does not:

- Grow GMO rice varieties.
- Use GMO farm inputs in the production of rice.
- Purchase GMO ingredients for their products.

Lundberg Family Farms does:

- Actively support keeping California a GMO rice free state.
- Support the consumers' right to know if a product contains GMO ingredients.

Since 1937, the Lundberg family has grown rice in the beautiful Sacramento Valley of Northern California. Today, Lundberg Family Farms is the largest producer of organic brown rice in the United States. The Lundbergs have taken the lead in protecting the state from genetically modified organisms. This statement was posted on the company's web site, www.lundberg.com on December 4, 1999.

Resources

PLANETARY HEALTH

- **Amberwaves**, a network of individuals, families, and organizations devoted to saving natural rice, wheat, and other essential foods from the threat of genetic engineering, with branches in different regions and countries. Box 487, Becket, MA 01223, fax (413) 623-6042. www.amberwaves.org (under construction).

INTERNET SITES

- **Ag Biotech InfoNet**, a web site devoted to biotechnology and genetic engineering, especially scientific reports and technical analysis, but covering many sides of the issue. www.biotech-info.net
- **Campaign to Label Genetically Engineered Foods**, a public advocacy group with comprehensive coverage of GE news. www.thecampaign.org
- **Yahoo GE Food Debate News Page** provides daily coverage of GE articles and news. www.yahoo.com/Full_Coverage/Science/Genetically_Modified_Food/

BOOKS AND ARTICLES

- Cummins, Ronnie and Ben Lilliston, *Genetically Engineered Food* (Marlowe & Co., 2000). Impassioned consumer's guide by leading activists.
- *Greenpeace True Food Shopping List* (Greenpeace, 2000). Comprehensive guide to non-GE foods, transitional foods, and GE foods in the marketplace.
- Ho, Mae-Wan, *Genetic Engineering: Dream or Nightmare* (Gateway Books, 1998). An impassioned critique by a scientist and a leader in the campaign against GE in England and around the world.
- Jack, Alex, *Imagine a World Without Monarch Butterflies* (One Peaceful World Press, 2000). An overview of the health and environmental effects of GE foods, including summaries of scores of scientific and medical studies.
- Shiva, Vandana, *Stolen Harvest: The Hijacking of the Global Food Supply* (South End Press, 2000). A Third World perspective on biopiracy.
- Stabinsky, Doreen, *California at the Crossroads* (Greenpeace, 2000). Special report on the impact of genetic engineering on California's agriculture. Available online: www.truefoodnow.org

GE-FREE FOOD COMPANIES

- **Eden Foods**, the pioneer natural foods manufacturer and distributor, has taken the lead in the campaign against GEOs and preserving organic purity. 701 Tecumseh Rd., Clinton MI 49236, (517) 456-7424. www.edenfoods.com
- **Lundberg Family Farms** is the largest producer of brown rice in the United States. P.O. Box 369, Richvale, CA 95974; (530) 882-4551. www.lundberg.com
- **Southern Brown Rice** is a leading grower and distributor of organic brown rice located in Arkansas. Box 165 Weiner, AR 72479; (800) 421-7423

The Amberwaves Network

“In the Far East, the ideogrammic word for ‘peace’—wa—is formed from the ideograms for ‘grain’ and ‘mouth.’ Ancient people intuitively knew that a diet based predominantly on grains and vegetables created a peaceful mind and society.”—MICHIO KUSHI AND ALEX JACK, ONE PEACEFUL WORLD

Humanity’s true wealth and heritage lie in the diversity of the earth’s natural plants and animals; its rich tapestry of cultures, culinary traditions, and customs; and the free flow of ideas, information, and seeds. The Amberwaves Network is being established to bring together farmers; food manufacturers, distributors, retailers, and consumers; health practitioners; environmentalists; religious and spiritual leaders; and individuals, families, and communities to protect and promote whole grains on our planet, focusing on rice as the signal crop to be preserved. The preliminary goals are as follows:

- **Crop Protection:** Declaration of the Sacramento Valley in California and rice-growing regions in Arkansas, Texas, and other areas, as well as restaurants, sushi bars, natural foods stores, supermarkets, churches, schools, hospitals, and other organizations across the country, as GE-Rice Free Zones.
- **Farm Protection:** Implementation of Identity Protection (IP), practical methods and safeguards to separate modified and conventional crops from seed to table; compensation for rice farmers, natural foods distributors, and retail suppliers whose crops or products are contaminated by GE rice in growing, storage, transportation, or processing; and protection for traditional rice strains that have been used for centuries from corporate patenting
- **Health and Environmental Protection:** A moratorium on planting, growing, distributing, or selling GE rice, pending comprehensive tests of its impacts on human health, the environment, organic farm-

ing, and the lives of hundreds of millions of farmers and their families who save natural rice seeds for planting each year

- **Consumer Protection:** Mandatory labeling of all GE rice and other foods in the marketplace; product liability to hold GE seed producers, growers, and brokers accountable for their products; and legal redress and health care treatment for consumers who may experience allergenic reactions, toxicity, or other harmful effects from ingesting GE rice

- **Rice Promotion:** Promotion of rice, especially organically grown brown rice, and other whole grains as the foundation of a healthy diet as recently recognized by the new *Dietary Guidelines for Americans*, 2000 edition, that calls upon the American people to consume brown rice and other whole grains daily

What Can I Do?

At a personal and community level, individuals, families, communities, businesses, and other organizations are encouraged to take the following practical steps:

- **Sign the Save Organic Rice Petition and Declaration of a GE-Rice Free Zone:** Sign the petition and declaration (on p. 80) and circulate them among friends, family members, and associates. Copies will be distributed to federal, state, and local officials; the United Nations, World Health Organization, and other international agencies; and to farm groups, the natural foods industry, supermarkets, restaurant chains, and other outlets, showing strong, grassroots opposition to GE rice.

- **Join Amberwaves:** By signing the petition you will automatically become a member of the Amberwaves Network and receive information on future activities.

- **Make a Donation:** Funds are urgently needed to print more literature, start a newsletter, open an office in the Sacramento Valley, and develop a web site. Amberwaves would also like to open offices in Washington, D.C., the United Nations, and send a delegation of organic farmers, environmentalists, and natural foods cooks to visit countries in Asia, Africa, and the Middle East Suggested donation: \$10/students and seniors; \$25/individuals; \$50/families and organizations; \$100/supporting donors; and \$1000/sustaining donors.

- **Sponsor a Speaker, Organize a Workshop, or Plan a Fundraising Dinner:** The Amberwaves Network is developing a speakers bureau of teachers, environmentalists, and farmers who can speak authoritatively on the issues.
- **Distribute Books and Literature:** Additional copies of *Saving Organic Rice* are available in bulk at discount. 2-4 copies: 20% off (\$5.56/each); 5-24 copies: 40% off (\$4.17/each); 25-49 copies, 45% off (\$3.82/each); and 50 or more: 50% off (\$3.48/each). Add \$4.95 s&h per order. Extra petitions, brochures, and other literature are also available inexpensively or at no cost.
- **Send Your Mailing List:** Amberwaves would like to mail literature to thousands of people in the natural foods, holistic, and environmental communities. If you have a computerized mailing list, please send us a copy on a floppy disk or Zip disk. You can prepare this easily as an export file from your database. Format as a “comma separated” file including separate fields (last name, first name, address, city, state, and zip code). Please indicate whether the disk is Mac or PC and the approximate number of names.
- **Volunteer Your Skills:** Help is needed in the following areas: administration, public relations, web development, publishing, public speaking, research, fundraising, farming/gardening, seed bank, etc.

AMBERWAVES
 P.O. Box 487
 Becket, MA 01223
 Fax (413) 623-6042

SAVE ORGANIC RICE PETITION

"I oppose genetically engineered rice and support efforts, including mandatory labeling of all GE foods, to preserve organic and natural rice, wheat, and other essential foods for my children, grandchildren, and future generations. Through my food choices, I will strive to keep America and the planet beautiful, healthy, and peaceful."

Signature_____Date_____

Name_____

Address_____

City/State/Zip_____

GE-RICE FREE ZONE DECLARATION

"I/we declare _____ a GE-Rice Free Zone and will not knowingly produce, sell, buy, or prepare genetically engineered rice or other foods." (Please circle locale: natural foods store, restaurant, school, church, farm, health center, or other_____)

Organization_____Contact Person_____

Address_____

City/State/Zip_____

Phone_____Email_____

Copies of the petition and declaration will be distributed to state and federal officials, farm organizations, food companies, the United Nations, and international agencies as part of the campaign to preserve organic rice and other staple foods. A GE-Rice Free Zone certificate is available for public display.

___ Send me additional petitions and other literature to circulate.

Enclosed is a donation to help preserve natural and organic rice, promote the benefits of whole grains, and create a healthy, peaceful planet.

___ \$10 ___ \$25 ___ \$50___ \$100 ___\$200 ___\$1000 _____ Other

Please make checks payable to: "Amberwaves." Amberwaves is a division of Planetary Health, Inc., a nonprofit educational organization.

Return to: AMBERWAVES
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“If current trends in the genetic engineering of California’s major export crops continue, the state’s producers could be facing . . . the potential collapse of its organic food industry.”

—GREENPEACE REPORT ON GE RICE & OTHER FOODS



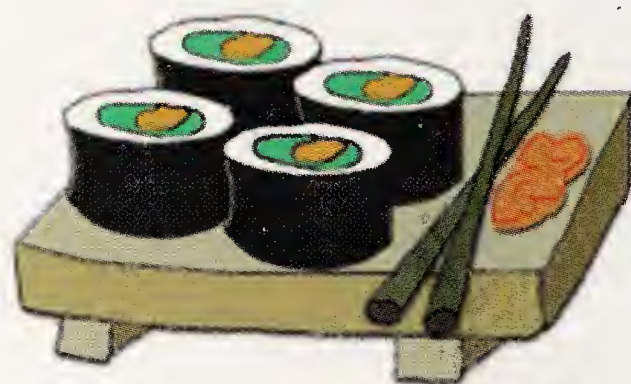
Three billion people—one half the world’s population—eat rice daily. The biotech industry claims that whole brown rice is nutritionally deficient and needs to be replaced with new “improved” genetically engineered varieties. GE rice boosts nutrients in the part of the grain that is retained after milling into 100% white rice. GE rice may cause allergies and produce other adverse health and environmental effects. Some GE rice may contain Bt, the deadly pesticide

that harms Monarch butterflies and other beneficial insects.

In the next few years, Monsanto and other biotech companies plan to convert half the nation’s rice to new GE varieties. Through cross-pollination, organic rice fields in California that produce 80% of the brown rice in America could become contaminated.

Meanwhile, no labeling is required, and consumers have no way of knowing whether their food contains GE ingredients. *Saving*

Organic Rice is a passionate critique of GE foods by leading scientists, environmentalists, natural health educators, and community activists. It offers practical steps to protect organic rice and other essential foods for future generations.



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Amberwaves

Leading the Campaign to Save Organic Rice
www.amberwaves.org

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Courtesy of Lundberg Family Farms*

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